LITERATURE MAP · STATUS 22 SEPTEMBER 2026

Condensates in twisted-bilayer optical lattices

Superfluidity across the moiré pocket crossover: how a weakly interacting condensate responds as the lowest band flattens

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Abstract. Twisted-bilayer optical lattices put two spin states of a Bose gas into square lattices rotated against each other and couple them with a microwave field, so that the Rabi coupling plays the part of interlayer tunnelling in twisted bilayer graphene. One experiment has been published, on ^{87}Rb at a twist of 5.21°; it reports two superfluid regimes and a superfluid–insulator transition that moves with the coupling. This literature map collects what is known about how such a condensate stays superfluid as its lowest band flattens. It brings together the experiment and the lattice and continuum theory written about it, the theory of superfluid weight and quantum geometry developed for magic-angle graphene, recent work on condensates in flat bands, and the measurements of the superfluid fraction in modulated Bose gases. For a square bilayer with uniform coupling there is no magic angle: the band flattens because the atoms localise in the AA registry pockets, and the bosonic flat-band theory then rules out a geometric superfluid weight. The map argues that the open physics lies in the crossover, where band mass, density inhomogeneity and Josephson tunnelling between pockets compete, and it lists the calculations that would settle it. Every number traces to a checked source or to the project's own single-particle band calculation, which is marked as such.

Keywords: twisted-bilayer optical lattice; moiré superlattice; Bose–Einstein condensate; superfluid fraction; quantum metric; flat band; Rabi coupling; Josephson array

At a glance

10	27	8	153	151	22
chapters	pages of text	figures	sources	sources checked	open questions

Status: complete draft, first edition. Literature status 21 September 2026; checks of sources, numbers and quotations carried out on 21–22 September 2026 with AI assistance; see "How this map was made" on page i.

The problem

Twisted-bilayer optical lattices bring moiré physics into a Bose gas. In the one experiment realised so far [1], two spin states of ^{87}Rb sit in square lattices rotated by 5.21° and a microwave field couples them, in the role that interlayer tunnelling plays in twisted bilayer graphene. The experiment reports a moiré density pattern, two distinct superfluid regimes and a superfluid–insulator transition that shifts with the coupling. The question is how the superfluidity of a weakly interacting condensate evolves as the lowest band of such a lattice flattens.

It matters because it tests, for bosons and without disorder, strain or Coulomb interaction, how far the quantum-geometric picture of magic-angle graphene carries over, on the only moiré quantum gas with published data. It is hard because in a square bilayer with uniform coupling the band flattens by localisation into AA registry pockets rather than by interference at a magic angle. In the crossover window $\Omega/J \approx 5$–25, where the experiment operates, conventional (m/m^*), geometric (condensate metric), density-inhomogeneity (Leggett) and inter-pocket Josephson contributions to the superfluid weight compete, and inter-pocket tunnelling can fall below accessible temperatures.

Established: uniform coupling gives no magic angle and acts as a scalar moiré potential proportional to Ω; a single flat band at a time-reversal-invariant momentum with uniform Bloch-state density cannot be superfluid for bosons within Bogoliubov theory; lattice models and Monte Carlo find clustered, glassy phases and thin superfluid domains destroyed at nK temperatures. Not yet found: Bogoliubov spectra, sound and the superfluid-fraction tensor of the Rabi-coupled condensate, its bosonic quantum geometry, finite temperature, and a trapped simulation compared with the data. Progress would be a quantitative account of superfluid fraction, sound and depletion across the crossover that reproduces the two superfluid regimes and the shifted transition.

How this map was made

Scope, method and conventions

Scope. Bose gases in twisted-bilayer optical lattices (experiment, lattice and continuum theory), flat-band bosons and quantum geometry, measurements and definitions of the superfluid fraction, the electronic moiré context, numerical methods and the research landscape; fermionic and photonic analogues only where they bear on the bosonic question.

Literature cutoff. 21 September 2026. Preprints are included and marked as such; published versions are cited where they were found.

How it was compiled. The starting documents were five project documents and a two-page proposal naming about 100 sources. Web searches extended them on twisted-bilayer optical-lattice theory, flat-band bosons, superfluid-fraction measurements, electronic moiré, numerical methods and the research landscape, 2011–2026. Every source was then looked up in a primary record (arXiv or ar5iv, publisher page, code repository) in ten checking rounds with AI assistance: of the 153 sources cited in this build, 51 were confirmed or corrected, 100 partly verified and 0 not verifiable.

Numbers. Each number and quotation traces to a line of the project's facts ledger. Quotations are verbatim from the checked source; numbers confirmed only in part carry "as reported" or "as quoted".

Conventions. $E_R = h \times 3.67\,\mathrm{kHz}$ for $a = 395\,\mathrm{nm}$ [1]; J single-layer nearest-neighbour hopping; Ω Rabi coupling (Ω_R in the experiment's notation); L_M moiré period; m commensurate index, $\tan(\theta/2) = 1/m$. The tag [computed for this map] marks numbers reproduced from the project's single-particle band-structure grid (no interactions, trap or temperature).

In one page: the field and the argument

Moiré physics reached quantum gases in 2023, when a ^{87}Rb condensate was loaded into two square lattices twisted by 5.21° and coupled by a microwave field [1]. The experiment found "two forms of superfluid and a modified superfluid to insulator transition". Magic-angle graphene suggests reading this through flat bands and quantum geometry [2–4]. We argue that for this lattice the reading has to change, in five steps.

1 **The lattice has no magic angle.** In twisted square bilayers flat bands need a spatially varying interlayer coupling with symmetry-enforced cancellation [5]; a uniform Rabi coupling instead acts on the lowest band as a scalar moiré potential proportional to Ω [6]. The band flattens because the atoms localise in the AA registry pockets: at $m = 8$, $V_0 = 8\,E_R$ the width falls to 0.25 of its uncoupled value at $\Omega/J \approx 16$ and to 0.004 at $\Omega/J \approx 32$, where the participation ratio is 0.008 and the Wannier spread 0.022 L_M^2 [computed for this map].

2 **Thesis 1: geometric superfluidity is excluded there.** For bosons, "There can therefore be no superfluidity in a single flat band at a TRIM where the Bloch state has uniform densities.", and stable two-dimensional flat-band superfluidity "requires at least three bands within Bogoliubov theory" [7] (preprint). A band of isolated pockets is the atomic limit, where "no current is possible." [4]. The same authors conclude that "flat-band superfluidity is much more difficult to achieve than flat-band superconductivity." [7]

3 **The open physics is the crossover.** Between a dispersive band and isolated pockets, band mass, density inhomogeneity (the Leggett bounds [8, 9]) and tunnelling between pockets all carry superfluid weight. No calculation of this competition for the Rabi-coupled condensate was found in the literature checked for this map. The tools to decide it exist in simpler geometries: sound anisotropy and Leggett bounds measured in modulated two-dimensional gases [10–12], and Josephson oscillations between the cells of a supersolid [13].

4 **Thesis 2, working hypothesis: SF-II is a Josephson array of AA pockets.** The experiment's second regime is described as "superfluid domains embedded in a gapped insulator" [1]. At $V_0 = 8\,E_R$ its moiré-visibility threshold $\Omega_R > 0.5\,E_R$ sits at $\Omega/J \approx 16$, where the lowest band detaches and has lost 75 % of its width [computed for this map]. The hypothesis fails if the superfluid fraction exceeds the Leggett prediction for the pocket density, if sound scales as U (the flat-band signature [14]), or if the threshold does not track Ω/J as V_0 changes.

5 **Progress** would be a two-component Gross–Pitaevskii calculation of the superfluid-fraction tensor, sound and depletion across the crossover, with Leggett brackets and a trap, that reproduces the two superfluid regimes and the shifted transition [1, 10, 12]. A finite-temperature treatment would show whether the inter-pocket tunnelling survives at the temperatures of the experiment, which the paper, as read for this map, does not state.

Where quantum geometry still enters

The geometric formulas do not disappear from the problem. The condensate metric at the condensation momentum sets sound and stability in the bosonic theory [7, 14], so it is a diagnostic of how far the crossover has progressed. Band touchings [15] and detuned couplings, which map onto Hofstadter-type spectra [16], are the places where a genuinely geometric window could still open. Chapter 3 explains these ideas for readers from the Bose-gas side.

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Terms from cold-atom physics, electronic moiré materials and quantum geometry, as this map uses them.

How to read this map

Twenty minutes. The problem, the one-page summary, then the key-points boxes at the end of each chapter.

New to electronic moiré and quantum geometry. Chapter 3, then chapter 6.

Looking for a research question. Chapters 6 and 9.

Boxes: key points, what a result means for the reader, open questions. Numbers tagged [computed for this map] come from the project's own band-structure grid.

1 The topic in brief

What a twisted-bilayer optical lattice is, what the one experiment saw, and which variants of the idea have been proposed.

Twisted bilayer graphene (TBG) is two graphene sheets rotated by a small angle and coupled by interlayer tunnelling; near a magic angle its lowest moiré bands become flat and the material turns superconducting [2, 3, 17]. A twisted-bilayer optical lattice (TBOL) copies the construction with cold atoms. The two "layers" are two internal states of the same atoms, each trapped in its own square lattice, the lattices are rotated against each other, and a microwave or Raman field couples the states in the place of interlayer tunnelling [1, 18]. Figure 1.1 follows the idea from graphene to what is measured; Table 1.1 lists the variants.

1.1 The platform

The one experiment so far used a Bose–Einstein condensate (BEC) of ^{87}Rb with "state $|1\rangle \equiv |F = 1, m_F = 1\rangle$ and state $|2\rangle \equiv |F = 2, m_F = 0\rangle$" [1]. Each state feels one square lattice: the two lattices are formed at "$\lambda_1 = 790.02\,\text{nm}$ and $\lambda_2 = 788.28\,\text{nm}$", both tune-out wavelengths, which a companion preprint studied with Kapitza–Dirac scattering [19]. The lattice constant is $a = 395\,\text{nm}$ and the recoil energy is $E_\text{R} = h \times 3.67\,\text{kHz}$; the experiment writes E_r, and we write E_R throughout. The twist is $\theta = 5.21°$ with an uncertainty of about $0.05°$, and the moiré period is measured to be $4.35\,\mu\text{m}$, eleven lattice constants. The condensate holds up to 5×10^5 atoms, and an accordion lattice with spacing tunable from 12 to $3\,\mu\text{m}$ compresses it into a pancake with "$\omega_z \approx 2\pi \times 1\,\text{kHz}$". A 6.8 GHz microwave couples the two states with a Rabi coupling Ω_R of up to $1\,E_\text{R}$; at $V_0 = 4\,E_\text{R}$ it was calibrated as $2\pi \times 893\,\text{Hz} = 0.243\,E_\text{R}$, or $\Omega_\text{R}/t = 2.07$ in units of the hopping t. After ramps of the lattices (30 ms) and the microwave (10 ms) and a hold of up to 40 ms, the gas is imaged in situ or after 18 ms of expansion [1]. The paper, as read for this map, does not state the temperature, the radial trap frequency or the atom number in the pancake.

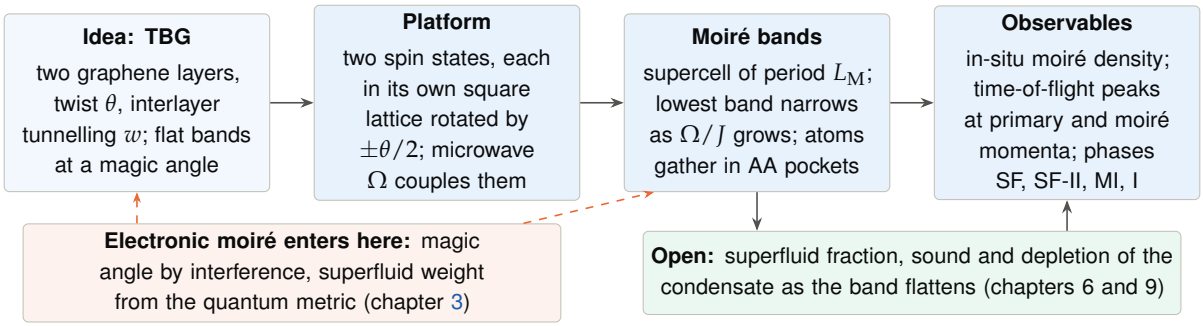


Figure 1.1: From twisted bilayer graphene to what a twisted-bilayer optical lattice measures. Top row: the chain of the platform [1]. Orange: where the electronic moiré ideas of chapter 3 are borrowed. Aqua: the quantity that links the bands to the observed phases and that has not been calculated for this platform in the literature checked for this map.

1.2 The four regimes

The paper reports "two forms of superfluid and a modified superfluid to insulator transition". Its labels are defined as follows: "SF, SF-II, MI, and I refer to superfluid, superfluid only with short-range coherence, Mott insulator, and insulator." [1] In SF, time of flight shows peaks both at the primary lattice momenta and at the moiré momenta. SF-II is described as "superfluid domains embedded in a gapped insulator"; as the project documents read the data, the moiré peaks are gone there while the primary peaks remain. In I the filling is integer but modulated

on the moiré scale. In the phase diagram, which extends to 24 E_R, along path I the coupling is fixed at $\Omega_R = 0.24\,E_R$ and the lattice is deepened, giving SF → SF-II → MI. Along path III the depth is fixed at 8 or 14 E_R and the coupling is raised; the moiré visibility "presents the threshold behavior and emerges at $\Omega_R > 0.5\,E_r$". The paper models the gas with a two-component Gross–Pitaevskii equation (GPE) with SU(2)-symmetric interactions $g_{11} = g_{22} = g_{12}$ and with a single-site Gutzwiller map of a Bose–Hubbard model, at the commensurate stand-in angle $2\arctan(1/22)$ and zero detuning [1]. A later experiment by the same group used an untwisted honeycomb lattice [20]; no second twisted-lattice experiment was found for this map.

1.3 What "flat band" means here

In TBG the band flattens by interference: tunnelling paths cancel while the states stay spread over the moiré cell (chapter 3). A uniform microwave coupling cannot do this in a square bilayer. Flat bands there need a spatially varying interlayer coupling with symmetry-enforced cancellation [5], and a uniform Ω acts on the lowest band as a scalar moiré potential proportional to Ω [6] (preprint). In the paper's own strong-coupling limit the lower dressed state feels $(V_1 + V_2)/2 - \Omega_R$, the average of the two lattices [1], which is deepest where their sites coincide, in the AA registry regions. A flat band here is therefore an array of weakly coupled AA pockets, the atomic limit, not a band held flat by interference.

Where the experiment sits on this scale is a question of Ω/J, with J the single-layer hopping. At $V_0 = 4\,E_R$, where according to the project documents the moiré and superfluid data were taken, $\Omega_R \leq 1\,E_R$ means $\Omega/J \leq 12$ and the lowest band keeps at least 93 % of its uncoupled width at $m = 8$ [computed for this map]; at $m = 12$ it keeps 59 %, and the experiment's angle, $m = 22$, was not computed. At $V_0 = 8\,E_R$ the path-III threshold $\Omega_R = 0.5\,E_R$ is $\Omega/J \approx 16$, where the band detaches from the next one and has lost 75 % of its width [computed for this map]. Chapter 2 gives the definitions and the numbers.

1.4 Variants of the platform

Table 1.1 sorts the proposals by how the moiré pattern enters. Most use two internal states with a uniform coupling, on square or honeycomb lattices. For square lattices Paul, Recher and Santos write the uniform coupling as an interlayer hopping that decays with the distance between sites and find Bose-glass-like "mobility islands" even at commensurate angles [21]; their earlier work found multifractal single-particle states [22]. With a detuning, the imbalanced limit "can be exactly mapped to Hofstadter's original model plus a long-range hopping term" [16]. On honeycomb lattices a uniform coupling does produce magic flat bands [23, 24]. Other variants drop the tunnelling and let interactions generate the moiré pattern [25, 26], sum two rotated lattices for a single component [27, 28], rotate one lattice continuously in time [29], or move to spin-1 condensates in twisted trilayer lattices [30] and Floquet-modulated interactions [31] (preprints). Photonic moiré lattices share the question of localisation in a moiré potential [32, 33].

Key points

- One experiment: ^{87}Rb, two spin states in square lattices twisted by 5.21°, microwave coupling up to 1 E_R, four regimes SF, SF-II, MI and I [1].
- Uniform coupling has no magic angle in a square bilayer; the band flattens by localisation in AA pockets [5, 6].
- Variants differ in how the moiré pattern enters; only one has data.

Table 1.1: Variants of the twisted-bilayer platform. "Uniform" means the coupling does not vary in space. Particles as stated in the sources or, where marked, in the project documents.

Variant	Coupling	Particles	What is studied	Sources
square lattices, uniform Ω	microwave (or Raman), uniform	bosons (^{87}Rb)	experiment: moiré density, two superfluids, shifted transition; lattice theory: clusters, mobility islands	[1, 18, 21]
honeycomb spin-twisted lattices	uniform Ω	fermions; atoms; fermions (docs)	magic flat bands for $\theta \lesssim 6°$; topological vs trivial flat bands; fractional Chern insulator (preprint)	[23, 24, 34]
moiré without a twist	spatially modulated Raman	fermions	quasi-flat bands with Dirac cones	[35]
interaction-induced moiré	none (contact interaction only)	bosons	induced and "interlocked" Mott insulators, Bose-glass-like phases, quenches	[25, 26]
summed twisted potential	one component, no coupling	bosons	continuous-space QMC: thin superfluid domains, Bose glass; tight-binding fits	[27, 28]
dipolar twisted bilayer	no tunnelling; dipolar interaction	hard-core bosons	QMC: insulating islands surrounded by superfluid at 5.21°	[36]
one-dimensional moiré	bichromatic lattice	bosons	moiré flat bands broadened by weak repulsion; "local supersolid" (preprint)	[37, 38]
photonic moiré	not applicable	photons	localisation and delocalisation of light; solitons	[32, 33]

2 Foundations

The geometry of the twisted supercell, the Rabi-coupled Gross–Pitaevskii model, how the lowest band turns into pockets, and the definitions of the superfluid fraction used in the rest of the map.

Numbers tagged [computed for this map] in this chapter come from the project's single-particle plane-wave bands at the commensurate index $m = 8$, without interactions, trap or temperature; the experiment's $m = 22$ was not computed.

2.1 Commensurate supercells and the moiré period

Two square lattices of constant a, rotated by $\pm\theta/2$, share a common supercell when $\tan(\theta/2) = 1/m$ for an integer m. The supercell is a square of side $\sqrt{m^2+1}\,a$ and holds m^2+1 sites of each layer; the moiré period is $L_\mathrm{M} = a/[2\sin(\theta/2)] = \sqrt{m^2+1}\,a/2$. At the experiment's $\theta = 5.21°$ this gives 4.35 μm, or 11 a, and the paper's theory uses the commensurate stand-in $\theta = 2\arctan(1/22) = 5.205°$, a supercell of 22 a [1]. The map's own numbers are for $m = 8$ ($\theta = 14.25°$, $L_\mathrm{M} = 4.03\,a$) and, in part, $m = 12$ ($\theta = 9.53°$, $L_\mathrm{M} = 6.02\,a$) [computed for this map]. Where the sites of the two lattices coincide we speak of AA registry, and where a site of one sits at a plaquette centre of the other, of AB registry (Fig. 2.1). Without coupling each layer's lowest band is folded into the small supercell Brillouin zone; at $m = 8$, $V_0 = 8\,E_\mathrm{R}$ the lowest folded band has a width $W_1(0) = 0.0347 \times 8J$ [computed for this map].

2.2 The Rabi-coupled Gross–Pitaevskii equation

The mean-field model of the experiment is a two-component GPE in which each component $\sigma = 1, 2$ feels its own lattice and the Rabi coupling converts one into the other ($\bar{\sigma}$ is the other component):

$$i\hbar\,\partial_t\psi_\sigma = \Big[-\frac{\hbar^2\nabla^2}{2m} + V_\sigma + V_T + \sum_{\sigma'} g_{\sigma\sigma'}|\psi_{\sigma'}|^2\Big]\psi_\sigma + \frac{\hbar\Omega}{2}\,\psi_{\bar{\sigma}}, \qquad V_\sigma(\mathbf{r}) = -V_0 \sum_{j=x,y} \cos^2(k_L\,\mathbf{e}_{\sigma j}\cdot\mathbf{r}). \tag{1}$$

Here $k_L = \pi/a$, the unit vectors $\mathbf{e}_{\sigma j}$ of lattice σ are rotated by $\pm\theta/2$, V_T is the trap, and we quote $\hbar\Omega$ in units of E_R and call it Ω. The experiment's model is quasi-two-dimensional with SU(2)-symmetric interactions $g_{11} = g_{22} = g_{12}$ [1]. Without interactions and trap, Eq. (1) is the 2×2 Hamiltonian whose bands the project computed. Without lattices a finite coupling "changes the first order demixing phase transition" of the mixture "to a second order" one [39]; with lattices it also mixes two different potentials. For strong coupling the lower dressed state sees $-(\hbar^2/2m)\nabla^2 + (V_1 + V_2)/2 - \Omega_\mathrm{R}$ [1]. The average of the two lattices is deepest at AA registry, and this is where the atoms collect.

2.3 A scalar moiré potential, and no magic angle

Wang, Gao and Shi project the coupled problem on the lowest band at small angle and obtain two decoupled branches in a moiré potential proportional to Ω [6]:

$$H_{\mathrm{eff},\pm} = \frac{p^2}{2m_\mathrm{eff}} \pm V_m(\mathbf{r}), \qquad V_m(\mathbf{r}) = 8\pi^{3/2}\,\Omega\,l_\mathrm{ho}^3 \sum_{\mathbf{G}} e^{-G^2 l_\mathrm{ho}^2}\, e^{i(R^{1/2}-R^{-1/2})\mathbf{G}\cdot\mathbf{r}}, \tag{2}$$

with l_ho the harmonic length of a lattice site, $\mathbf{G}$ the reciprocal lattice vectors and R the rotation by θ. The derivation is written for a three-dimensional quasi-moiré crystal and holds "When the twist angle is small such that $\theta \cdot l_\mathrm{ho}^{-1} \ll 1$"; applying it to the two-dimensional square bilayer is a step taken by the project, not by the preprint. The amplitude grows linearly with Ω and carries a Gaussian form factor, and nothing in Eq. (2) singles out an angle. Kariyado and Vishwanath reach the same conclusion from symmetry: in twisted Bravais bilayers, flat bands need a spatially varying interlayer coupling whose matrix elements vanish for particular momenta and shifts, "an interference effect where symmetry forces exact cancellation", and a uniform coupling lacks this (our paraphrase) [5]. In the lattice language of Paul, Recher and Santos the same coupling becomes an interlayer hopping $t_\perp e^{-|\mathbf{R}_{1j}-\mathbf{R}_{2j'}|^2/4l_0^2}$ that is largest between coinciding sites [21].

2.4 Pocket formation

The lowest band flattens once the potential localises the atoms at its minima (Fig. 2.2; all at $m = 8$, $V_0 = 8\,E_\mathrm{R}$) [computed for this map]. At $\Omega/J = 3.2$ the band keeps 98 % of its folded width, the band-mass ratio at Γ is $m/m^* = 0.28$, the participation ratio (PR) of the band-bottom density is 0.38 and the coarse-grained AA/AB density ratio 2.4. At $\Omega/J = 16.2$ the width is $0.25\,W_1(0)$, a gap of $0.0007\,E_\mathrm{R}$ has opened to the second band, $m/m^* = 0.12$, PR = 0.059 and AA/AB = 110; at $\Omega/J = 32.4$ the four numbers are 0.004, 0.003, 0.008 and 3.7×10^3, with a gap of $0.010\,E_\mathrm{R}$. The width halves at $\Omega/J \approx 12$ for $V_0 = 8\,E_\mathrm{R}$, at ≈ 9 for $12\,E_\mathrm{R}$ and at ≈ 35 for $4\,E_\mathrm{R}$ [computed for this map], so the threshold is set by Ω/J with a depth-dependent prefactor, not by the angle.

The dependence on m is not settled: at $V_0 = 4\,E_\mathrm{R}$, $\Omega = 1\,E_\mathrm{R}$ the width ratio is 1.02 ($m = 6$), 0.93 ($m = 8$) and 0.59 ($m = 12$) [computed for this map], so the project documents' claim of a nearly angle-independent threshold for $m \geq 6$ fails at $4\,E_\mathrm{R}$, and $m = 22$ was not computed.

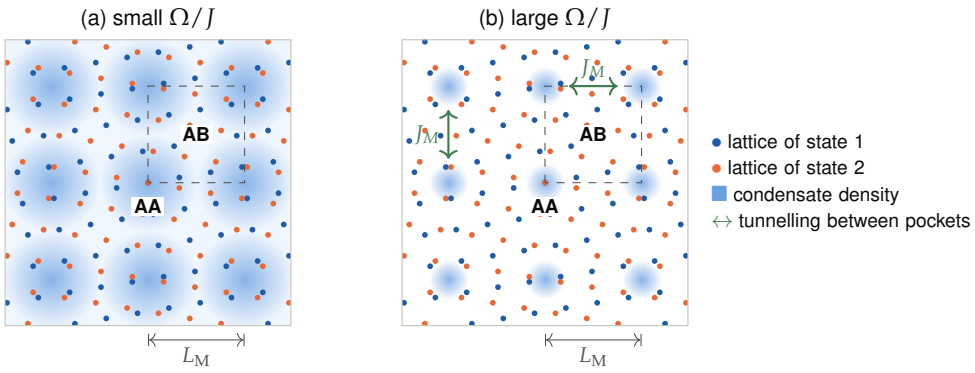
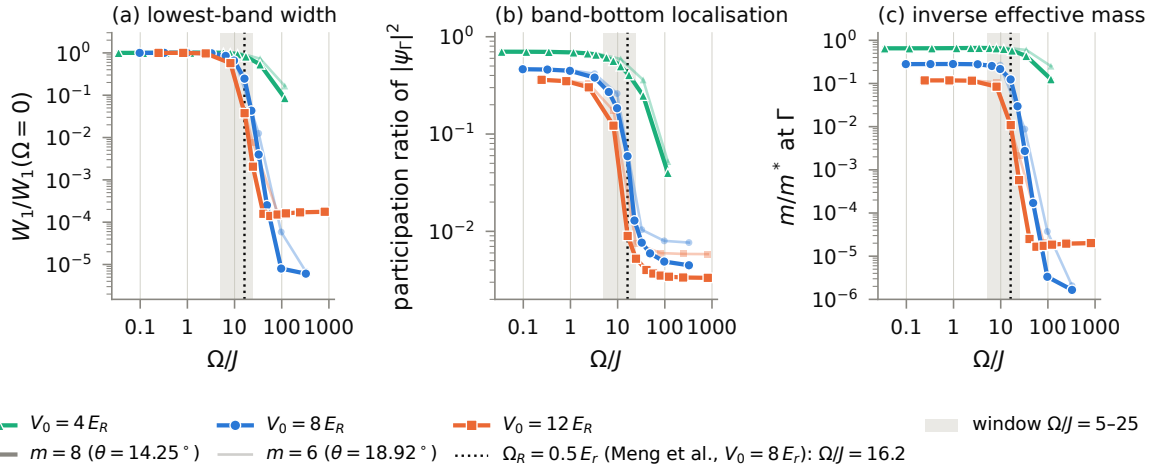


Figure 2.1: Schematic of pocket formation (not computed; drawn at $m = 6$, $\theta = 18.9°$, for visibility). Dots: the two rotated square lattices, which coincide at AA registry and alternate at AB. (a) For small Ω/J the condensate density is spread over the moiré cell with a mild AA excess. (b) For large Ω/J it sits in AA pockets connected by a weak tunnelling J_M, and the lowest band is an array of nearly isolated pockets.



single-particle bands; no interactions, trap or temperature; computed for this map from grid.csv

Figure 2.2: Pocket crossover in the single-particle bands [computed for this map]: (a) width of the lowest band relative to its folded value, (b) participation ratio of the band-bottom state, (c) band-mass ratio m/m^* at Γ, against Ω/J for $V_0 = 4$, 8 and 12 E_R at $m = 8$ (faint: $m = 6$). Shaded: the window $\Omega/J = 5$–25; dotted: the experiment's moiré-visibility threshold $\Omega_R = 0.5\,E_R$ at $V_0 = 8\,E_R$ [1], $\Omega/J = 16.2$. No interactions, trap or temperature; the experiment's $m = 22$ was not computed.

2.5 Superfluid fraction

The superfluid fraction f_s is the part of the gas that does not follow a slow motion of the potential. In a phase-twist calculation one imposes a velocity v and reads f_s off the energy, $E(v) = E(0) + f_s N m v^2/2$ for $v \to 0$ [9]; in an anisotropic lattice f_s is a tensor [40]. Leggett bounded f_s from above by the density modulation alone [8] and defined it for an arbitrary many-body system at zero temperature [41]. In two dimensions, for a twist along x, Pérez-Cruz, Astrakharchik and Massignan give the pair of bounds

$$f^+ = \frac{1}{\bar{n}} \frac{1}{\langle 1/\langle n\rangle_{r_\perp}\rangle_x}, \qquad f^- = \frac{1}{\bar{n}} \left\langle \frac{1}{\langle 1/n\rangle_x} \right\rangle_{r_\perp}, \tag{3}$$

where $\langle\cdot\rangle_x$ and $\langle\cdot\rangle_{r_\perp}$ average along and across the twist, and they find that "The bracketing is tighter for a 2D optical potential than for a grid of columns" [9]. A later preprint studies

the angular dependence in composite potentials [42]. Experiments use sound: with c_x and c_y the sound velocities along and across a one-dimensional modulation, $f_{s,x} = c_x^2/c_y^2$ and $c_x^2 = f_{s,x}/(m\kappa)$ with κ the compressibility, and for a weakly interacting condensate in a one-dimensional periodic potential Leggett's bound becomes an identity [10], and in a Bragg-phonon measurement "the speed of sound saturates Leggett's bound" [11]. For a condensate in one Bloch state, Taylor and Zaremba derive the sound speed $s = \sqrt{(\bar{n}/m^*)(\partial\mu/\partial\bar{n})}$ with $(m^*)^{-1} = \hbar^{-2}\partial^2\tilde{\varepsilon}/\partial k^2$ the inverse band mass [43]. Comparing with the sound relation above gives $f_s = m/m^*$, the relation we use for a Bloch condensate; it is our combination of the two results. The computed m/m^* is a single-particle number and only an estimate of f_s once interactions smooth the density.

2.6 The Bose–Hubbard mapping and where it fails

Lattice models replace Eq. (1) by a Bose–Hubbard model with hopping J and on-site interaction U; the experiment quotes $U/t = 1.67$ [1]. The exact single-layer hopping, taken from the one-dimensional Mathieu band, is $J = 0.0862$, 0.0308 and 0.01225 E_R at $V_0 = 4$, 8 and 12 E_R [computed for this map]. The asymptotic formula used in the project code, $J \approx (4/\sqrt{\pi})\, V_0^{3/4} e^{-2\sqrt{V_0}}\, E_R$ with V_0 in units of E_R, from the deep-lattice limit discussed by Zwerger [44], is too large by 36, 22 and 16 % at these depths [computed for this map], so every Ω/J in the literature depends on which J was used. The single-band description itself has limits: in continuous-space quantum Monte Carlo (QMC) for hexagonal potentials the bands match tight binding for $V_0 \gtrsim 5\,E_R$, yet at 15 E_R "the Bose–Hubbard approximation fails to describe but the first Mott lobe", with "strong density-assisted tunneling effects beyond the Bose-Hubbard framework" [45].

2.7 Energy scales

With $E_R = h \times 3.67\,\mathrm{kHz}$ for ^{87}Rb at $a = 395\,\mathrm{nm}$ [1], the experiment's couplings $\Omega_R = 0.24$–$1\,E_R$ at $4\,E_R$ correspond to $\Omega/J = 2.8$–11.6 [computed for this map]. Between pockets the tunnelling can be much smaller. For single-component bosons in a summed twisted potential, Johnstone *et al.* ran continuous-space worm QMC at $k_BT = 0.01$–$0.04\,E_R$ in boxes 5 and 15 moiré periods wide. They report maximal hoppings in the thin superfluid domains between 10^{-11} and $10^{-3}\,E_R$, and find that "thermal broadening strongly depletes the SF domains" [27]; the experiment's temperature is not stated. Measured scattering lengths exist for a different pair of ^{87}Rb states, $|F = 1, m_F = -1\rangle$ and $|F = 2, m_F = +1\rangle$: $a_{12} = 98.006(16)\,a_0$ and $a_{22} = 95.44(7)\,a_0$ [46]. That paper cites earlier values, $a_{12} = 97.66\,a_0$ and $a_{22} = 95.0\,a_0$ as quoted, which the project documents attribute to [47]. Values for the experiment's pair $|1,1\rangle$, $|2,0\rangle$ were not found for this map.

Key points

- Commensurate supercells: $\tan(\theta/2) = 1/m$, side $\sqrt{m^2+1}\,a$, $L_M = a/[2\sin(\theta/2)]$; the experiment is $m = 22$, the map's numbers are $m = 8$.
- A uniform Ω acts as a scalar moiré potential $\propto \Omega$; the band flattens by localisation in AA pockets, with its width halved at $\Omega/J \approx 12$ for 8 E_R [computed for this map].
- The superfluid fraction has three working definitions: phase twist, Leggett brackets and sound anisotropy; for a Bloch condensate $f_s = m/m^*$.
- Exact and asymptotic J differ by 16–36 %, and inter-pocket tunnelling can fall far below J.

3 Context: electronic moiré and quantum geometry for Bose-gas physicists

How flat moiré bands in graphene become superconducting through quantum geometry, and which parts of that account survive for a condensate in a twisted optical lattice.

We borrow most of the vocabulary for twisted-bilayer optical lattices (TBOL) from twisted bilayer graphene (TBG). This chapter covers how a moiré band becomes flat, why a flat band can still carry superfluid weight, and what changes for a condensate. Figure 3.1 traces the argument; Table 3.1 translates the terms.

3.1 Flat bands by interference: the magic angle

In the Bistritzer–MacDonald (BM) continuum model the Dirac cones of the two graphene layers are coupled by an interlayer tunnelling w that varies with the moiré pattern [2]. One dimensionless number controls the low-energy bands, $\alpha = w/(vk_\theta)$, where v is the Dirac velocity and k_θ the moiré wavevector. The renormalised velocity is $v^*/v = (1-3\alpha^2)/(1+6\alpha^2)$, and in the authors' words, "For a discrete set of magic angles the velocity vanishes, the lowest moire band flattens". The first magic angle is $\theta \approx 1.05°$. Nothing localises here. Tunnelling paths interfere destructively while the states stay extended over the moiré cell, and this is exactly where the optical lattice will differ.

Experiments then found correlated insulators at half filling of the flat bands [48] and superconductivity at "twist angles of about 1.1°—the first 'magic' angle", with "a critical temperature of up to 1.7 kelvin" at "a carrier density of about 10^{11} per square centimetre" [3, 17].

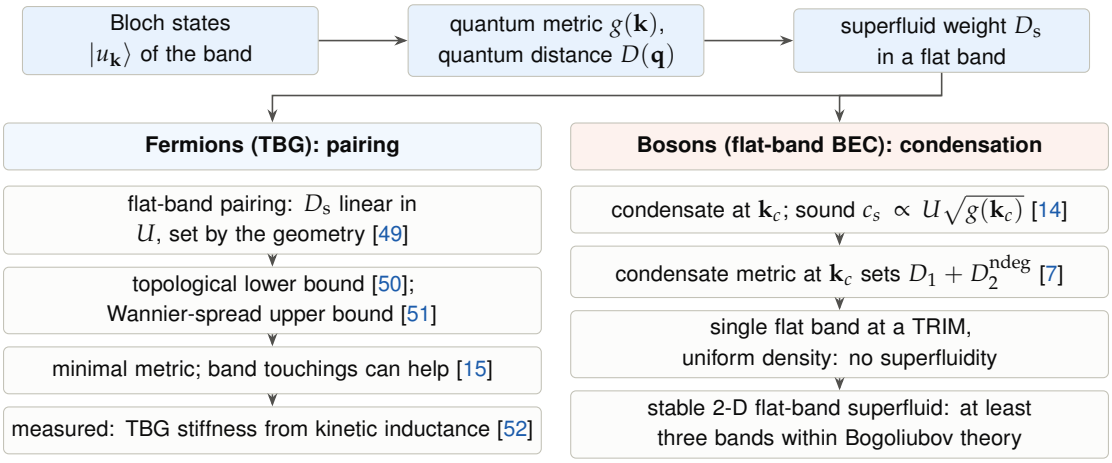


Figure 3.1: From Bloch states to superfluid weight: the shared trunk (top), then the fermion branch (pairing) and the boson branch (condensation at $\mathbf{k}_c$; [14] and the preprint [7]).

3.2 Superfluid weight from quantum geometry

In a dispersive band $D_s \propto n/m^*$, so a flat band with $m^* \to \infty$ looks like a dead end. It is not. The quantum metric $g(\mathbf{k})$ measures how fast the Bloch state $|u_\mathbf{k}\rangle$ changes with $\mathbf{k}$, and the quantum distance $1 - |\langle u_\mathbf{k}|u_{\mathbf{k}'}\rangle|^2$ is its finite-step version. Peotta and Törmä showed that pairing in a flat band still produces a superfluid weight, linear in the attraction ("$D_s \propto U$ in a flat band", as the abstract puts it) and controlled by this geometry [49]. For TBG the superfluid weight has "a topological lower bound proportional to the pairing gap" [50]. From the other side, Verma, Hazra and Randeria bound the low-energy optical weight, and so the stiffness, of a trivial flat

band by the Wannier spread [51]: well-localised orbitals mean little stiffness. Huhtinen *et al.* then showed that the physically relevant quantity is the minimal quantum metric, which "is related to the superfluid weight in isolated flat bands", and that "a non-isolated flat band can actually be beneficial" [15]. The review [4] states the limiting case plainly: "a trivial flat band is equivalent to an insulator in the atomic limit of large atom-atom separation", and there "no current is possible."

The stiffness of magic-angle TBG has now been measured. Tanaka *et al.* combined "d.c. transport and microwave circuit quantum electrodynamics to directly measure the superfluid stiffness of superconducting MATBG through its kinetic inductance" and found it "much larger than expected from conventional Fermi liquid theory" [52]; a companion measurement did the same in twisted trilayer graphene [53, 54]. A BM-model decomposition attributes 22–26 % of D_s at charge neutrality to quantum geometry [55] (preprint).

3.3 Condensates in flat bands

A condensate has no pairing gap. It occupies one Bloch state at momentum $\mathbf{k}_c$, and in a flat band the interaction energy Un_0 is the only scale. Julku, Bruun and Törmä worked this out for the kagome lattice [14]. The projected Bogoliubov spectrum is

$$E_1(\mathbf{k}_c+\mathbf{q}) = \frac{Un_0}{3} D(\mathbf{q}), \qquad D(\mathbf{q}) = \sqrt{1-|\langle u_1(\mathbf{k}_c+\mathbf{q})|u_1(\mathbf{k}_c-\mathbf{q})\rangle|^2}, \tag{4}$$

so the excitations are set by the quantum distance, the sound velocity $c_s = (2Un_0/3)\sqrt{g(\mathbf{k}_c)}$ is linear in U rather than in $\sqrt{U}$, and the depletion as $U \to 0$ "is determined by the quantum geometry only". The companion general theory [56] assumes, as the project documents summarise it (the paper was not read for this map), a uniform condensate density over the orbitals, an isolated band and projected interactions. Julku, Salerno and Törmä added "orbital-position-independent generalizations of the quantum metric" and showed that D_s must be taken at fixed total particle number [57].

The 2026 preprint by Huhtinen *et al.* sharpens the picture [7]. For uniform-density condensates at weak coupling, $D_1 + D_2^{\mathrm{ndeg}} = (2Un_0^2/M)\,\mathcal{M}^D(\mathbf{k}_c)$, where $\mathcal{M}^D$ is the condensate quantum metric: a local quantity at $\mathbf{k}_c$, not an integral over the Brillouin zone (BZ). The global geometry enters the fluctuation term D_3', which in their model "is positive only at very small interactions, and turns negative around $U \approx 0.13t$". At a time-reversal-invariant momentum (TRIM), "There can therefore be no superfluidity in a single flat band at a TRIM where the Bloch state has uniform densities." And a large integrated metric does not help: on the checkerboard lattice, "even though the integrated quantum metric diverges due to the band touching, condensation and superfluidity remain impossible." The authors conclude that stable two-dimensional flat-band superfluidity "requires at least three bands within Bogoliubov theory" and that "flat-band superfluidity is much more difficult to achieve than flat-band superconductivity." A companion preprint phrases the criterion in real space: "frameworks consisting of triangles with nonzero area are promising for condensation, whereas … square frameworks indicate condensation in a single mode is impossible" [58].

Two further preprints add to this. For a two-band condensate, "in the case of U(2) invariant interactions, the superfluid density is zero" [59]; the TBOL experiment is modelled with SU(2)-symmetric interactions $g_{11} = g_{22} = g_{12}$ [1], but no source checked tests the result there. Burgher *et al.* relate the sound velocity in quasi-flat bands to a "quantum geometric kernel" [60].

Table 3.1: Dictionary between twisted bilayer graphene and twisted-bilayer optical lattices.

TBG	TBOL	What it means
layer	spin state, each with its own lattice [1]	the "layers" are internal states, not planes in space
interlayer tunnelling w	Rabi coupling Ω (microwave)	uniform in space for a microwave drive
twist angle, magic angle	twist θ, index m; no magic angle for uniform Ω [5]	flattening is set by Ω/J, not by θ
AA / AB stacking	AA / AB registry regions	where the two lattices' sites coincide or alternate
BM continuum model	scalar moiré potential $\propto \Omega$ [6]	one effective band in a moiré potential
superfluid weight D_s	superfluid fraction f_s (a tensor)	response to a phase twist
quantum metric	condensate quantum metric at $\mathbf{k}_c$ [7]	local geometry, not the BZ integral
Cooper pairs	condensate in one Bloch state	what the superfluid is made of
Coulomb interaction	contact interaction g	short range, tunable, no screening
transport, kinetic inductance [52]	sound velocity, phase twist, Leggett bounds [10, 11]	how the stiffness is measured

3.4 What carries over, and what does not

Both systems ask what carries superfluid weight once the kinetic energy of a moiré band is gone. TBG samples come with twist-angle disorder, strain and long-range Coulomb repulsion, and the superfluid is made of pairs. The optical lattice is clean, the interaction is a contact term, and the superfluid is one macroscopic state whose density can be imaged; its superfluid fraction follows from sound anisotropy, $f_{s,x} = c_x^2/c_y^2$, and from Leggett's bounds on that density [10–12].

The real difference lies in how the band flattens. Flat bands in twisted square bilayers need a spatially varying interlayer coupling whose symmetry forces exact cancellations [5]. A microwave Rabi coupling is uniform, and projected on the lowest band it acts as a scalar moiré potential of amplitude proportional to Ω (derived for 3-D quasi-moiré crystals and applied to 2-D by the project) [6]. The band flattens by localisation in the AA pockets, not by interference, and at $m = 8$, $V_0 = 8\,E_R$ the coarse-grained AA/AB density ratio is already 110 at $\Omega/J \approx 16$ [computed for this map]. The bosonic geometric results assume the opposite, a uniform density.

What this means for Bose-gas physicists

A flat-band condensate has a clear signature: sound linear in U and a depletion that stays finite as $U \to 0$ [14]. The metric that counts is the local one at $\mathbf{k}_c$, not the BZ integral of the fermionic literature. In a square lattice with uniform coupling the flat limit is the atomic limit, so the geometric formulas become diagnostics of the crossover, not a mechanism.

Open

No calculation of the condensate metric of the Rabi-coupled bands was found for this map. Open: what the uniform-density results say for AA pockets, whether [59] transfers, and how the geometric term compares with the Leggett bound.

Key points

- TBG flattens by interference, a uniform-Ω TBOL by localisation [2, 5, 6].
- Bosons: one flat band at a TRIM with uniform density is not superfluid [7].
- AA pockets break uniform density: the open physics is in the crossover.

4 Data, benchmarks and verification

What the one experiment measures, how a superfluid fraction is measured and computed in modulated Bose gases, and which cross-checks and pitfalls a calculation for twisted-bilayer lattices has to respect.

4.1 How the experiment identifies its phases

Meng *et al.* use two probes [1]. In situ, at 50× magnification, they resolve the moiré density pattern with a period of 4.35 μm, eleven lattice constants of $a = 395$ nm. After 18 ms of time of flight (TOF), imaged state by state, the momentum distribution shows peaks at the primary lattice momentum and at the moiré momentum; with a moiré period of $11a$ the second set sits at 1/11 of the first. The phases are defined by the visibilities of the two peak sets: the superfluid (SF) shows both. The second regime, SF-II, keeps the primary peaks and loses the moiré ones; the figure caption calls it "superfluid only with short-range coherence", and the Extended Data describe "superfluid domains embedded in a gapped insulator". The Mott insulator (MI) and a moiré-modulated insulator (I) complete the diagram. Along path I the coupling is held at $\Omega_R = 0.24\,E_\mathrm{R}$ and the depth is raised (SF → SF-II → MI). Along path III the coupling rises at a depth in the Mott region, and "the visibility at the moiré momenta presents the threshold behavior and emerges at $\Omega_R > 0.5\,E_r$". The experiment writes E_r; we write E_R throughout.

The numeric phase-diagram anchor points are in the Extended Data of the Nature version only, not in the arXiv PDF, and this map could not confirm them; temperature, radial trap frequency and pancake atom number were not found in the pages read.

4.2 How the superfluid fraction is measured

Three groups have measured f_s in modulated Bose gases since 2023 (Table 4.1). Chauveau *et al.* imposed a 1-D modulation of period $d = 3.93(4)$ μm and amplitude up to 80 nK on a 2-D ^{87}Rb gas and compared Leggett's formula on the measured density with the sound anisotropy $f_{s,x} = c_x^2/c_y^2$; for a 1-D modulation of a weakly interacting condensate "the bound reduces to an identity" [10]. Tao, Zhao and Spielman Bragg-excited phonons in a 1-D lattice of period $a = 266$ nm (depth up to 10 E_R, healing length $\xi = 170(20)$ nm, $N \approx 2 \times 10^5$) and found that "the speed of sound saturates Leggett's bound" [11]. Rabec *et al.* went to a 2-D triangular lattice ($d = 6.0(1)$ μm, $\mu_0/k_B = 45(2)$ nK, $T < 20$ nK, depths up to 4.7 μ_0) and obtained the tensor by a static route (density profile and continuity equation) and a dynamic one (compressibility and sound); both agree with Gross–Pitaevskii equation (GPE) simulations and with the Leggett bounds from the measured profiles [12]. In a dipolar supersolid, Biagioni *et al.* read f_s off the Josephson oscillations of single cells and found "a relatively large sub-unity superfluid fraction" [13]. Preprints test Leggett's bound for strongly interacting bosons (not checked for this map) [61] and propose vortex necklaces as a probe [62].

Table 4.1: Measured superfluid fractions in modulated Bose gases; none is moiré or Rabi-coupled.

Group / platform	Modulation	Period	Method	Compared with
Dalibard/Beugnon, 2-D ^{87}Rb [10]	1-D, $V_0 \leq 80$ nK	3.93(4) μm	density + Leggett; c_x^2/c_y^2	Leggett bound (exact in 1-D)
Spielman, ^{87}Rb [11]	1-D lattice, $\leq 10\,E_\mathrm{R}$	266 nm	Bragg phonons, sound speed	Leggett bound, GPE
Dalibard/Beugnon, 2-D ^{87}Rb [12]	triangular, $\leq 4.7\,\mu_0$	6.0(1) μm	continuity eq.; compressibility + sound	GPE, Leggett bounds
Modugno, dipolar supersolid [13]	self-organised	–	Josephson oscillations of cells	sub-unity f_s

4.3 How the superfluid fraction is computed

Four routes are in use. The phase twist gives $E(v) = E(0) + f_s N m v^2/2$ for $v \to 0$ [9, 41]. Leggett's brackets give an upper bound f^+ and an explicit lower bound f^- from the density alone, and "The bracketing is tighter for a 2D optical potential than for a grid of columns" [8, 9]; a 2026 preprint studies their angular dependence in composite potentials [42]. Sound gives $c_x^2 = f_s/(m\kappa)$ [10] and, in a lattice, $s = \sqrt{(\bar{n}/m^*)\,\partial\mu/\partial\bar{n}}$ [43] [63]. The band route links nonclassical translational inertia to the effective mass, with f_s a rank-2 tensor [40]. The unitary Fermi gas has its own Leggett test [64]. Quantum Monte Carlo (QMC) takes $f_s = \Upsilon/\rho$ from winding numbers [27, 65].

4.4 Cross-method checks

For the square Bose–Hubbard model, cluster Gutzwiller extrapolated from clusters of up to 4×4 sites gives $(J/U)_c = 0.0596(4)$ against the QMC value 0.0597 [66, 67]; the single-site Gutzwiller of the experiment's theory has no such check for the moiré case. Continuous-space QMC shows where the lattice model ends: the bands match tight binding for $V_0 \gtrsim 5\,E_R$, but at $15\,E_R$ "the Bose–Hubbard approximation fails to describe but the first Mott lobe", with density-assisted tunnelling beyond it [45]. The Gaussian-overlap interlayer coupling of the two-layer tight-binding models [18, 21] reproduces the lowest bandwidth to 5–8 % for $\Omega/J \leq 3$, is 30 % low at $\Omega/J \approx 10$ and $10\times$ low at $\Omega/J \approx 32$ ($m = 8$, $V_0 = 8\,E_R$; the underlying table was not re-read for this map) [computed for this map]. Neural quantum states (20×20) [68] and infinite projected entangled-pair states (iPEPS) [69] give variational references.

4.5 Pitfalls

Classical-field methods need "the number of field modes in the Wigner simulation [to] be smaller than the number of particles" [70] and "all c-field region modes" to be "highly occupied" [71–73]. Near Mott lobes the truncated Wigner approximation (TWA) is fragile. For the SU(3) discrete TWA [74], Nagao and Yunoki report a breakdown for $t \gtrsim 0.1\,\hbar/J_{\mathrm{fin}}$ in strongly interacting 2-D Bose–Hubbard quenches and deviations from experiment "even at relatively short times"; their cluster version uses 2×2 clusters on 20×20 sites [75]. Kibble–Zurek (KZ) exponents mislead: 0.31 ± 0.03 against the KZ value 3/4 in a 3-D ^{87}Rb lattice [76]; exponents well below the KZ value and dependent on the final interaction for ^{39}K [77]; and iPEPS needs bond dimension $D = 11$–14 at a cost $\propto D^8$ to test the KZ prediction $\hat{\xi} \propto \tau_Q^{0.40}$ [78]. Energy scales can vanish: in twisted single-component potentials superfluid domains carry tunnelling $J_{\max}$ from 10^{-11} to $10^{-3}\,E_R$ and thermal broadening at $k_B T = 0.01$–$0.04\,E_R$ "strongly depletes" them [27].

Key points

- Phases are defined by TOF visibilities at two momenta; numeric anchors only in the Nature Extended Data [1].
- f_s is measured via Leggett bounds, sound and Josephson oscillations, never yet in a moiré gas [10–13].
- Gaussian-overlap couplings fail beyond $\Omega/J \approx 10$ [computed for this map]; TWA near Mott lobes, KZ exponents and nK tunnelling trap dynamics [27, 75, 76].

5 State of the art

One experiment, about two dozen theory papers written around it or its relatives, and a clear pattern in what they treat: single-particle bands and ground states, rarely excitations, temperature or dynamics.

This chapter first fixes the experiment, then sorts the theory by approach. Figure 5.1 shows which study treats which ingredient, Figure 5.2 places the work in time, and the last subsection lists what can be downloaded.

5.1 The experiment

Meng *et al.* loaded a ^{87}Rb Bose–Einstein condensate (BEC) with "up to 5×10^5 atoms" into two square lattices of constant $a = 395\,\mathrm{nm}$, each felt by one spin state, $|1\rangle = |F{=}1, m_F{=}1\rangle$ and $|2\rangle = |F{=}2, m_F{=}0\rangle$ [1]. The lattice beams sit at the tune-out wavelengths 790.02 and 788.28 nm, measured by Kapitza–Dirac scattering in a companion metrology paper [19]. The recoil energy is $E_\mathrm{R} = h \times 3.67\,\mathrm{kHz}$ (written E_r in the paper). The twist is $\theta = 5.21^\circ$ with an uncertainty of about 0.05°, the measured moiré period $4.35\,\mu\mathrm{m}$, and the paper's theory uses the commensurate angle $\theta = 2\arctan(1/22)$. An accordion lattice at 532 nm, tunable from 12 to $3\,\mu$m, makes the gas quasi-2-D with $\omega_z \approx 2\pi \times 1\,\mathrm{kHz}$. A 6.8 GHz microwave couples the layers with Ω_R up to $1\,E_\mathrm{R}$; at $4\,E_\mathrm{R}$ the calibrated Rabi frequency is $2\pi \times 893\,\mathrm{Hz}$, $\Omega_R/t = 2.07$, and the paper quotes $U/t = 1.67$. The Rabi oscillations decay at $1200\,\mathrm{s}^{-1}$. The sequence ramps both lattices in 30 ms, then the microwave in 10 ms, and holds for up to 40 ms. The phase diagram spans depths up to $24\,E_\mathrm{R}$. The paper's theory consists of a continuum band calculation with the strong-coupling limit $H^-_\mathrm{eff} \approx -(\hbar^2/2m)\nabla^2 + (V_1 + V_2)/2 - \Omega_R$, a two-component GPE with SU(2) interactions $g = g_{11} = g_{22} = g_{12}$, and a single-site Gutzwiller map. The headline is "two forms of superfluid and a modified superfluid to insulator transition". No second twisted-lattice experiment was found for this map; the same group has since studied a vortex at the Dirac point of an untwisted honeycomb lattice [20].

5.2 Single-particle continuum theory

Five results fix the band structure. First, the preprint by Wang, Gao and Shi projects the Rabi-coupled Hamiltonian on the lowest band at small angle and obtains a scalar moiré potential $V_m(\mathbf{r}) = 8\pi^{3/2}\Omega\, l_\mathrm{ho}^3 \sum_\mathbf{G} e^{-G^2 l_\mathrm{ho}^2} e^{i(R^{1/2} - R^{-1/2})\mathbf{G}\cdot\mathbf{r}}$, derived for a 3-D quasi-moiré crystal and valid for $\theta\, l_\mathrm{ho}^{-1} \ll 1$ [6]; its amplitude is proportional to Ω times a Gaussian form factor. Second, with a detuning δ, Wan, Gao and Shi find in the imbalanced limit ($\delta \gg V_0, \Omega$) an on-site term $v = \Omega^2 V_0/\delta^2$ that "can be exactly mapped to Hofstadter's original model plus a long-range hopping term", with the twist playing the flux and fractal spectra as a result [16]. Third, Kariyado and Vishwanath showed that flat bands in twisted Bravais bilayers come from a spatially varying interlayer coupling whose symmetry forces exact cancellations; a uniform coupling lacks that mechanism [5]. Fourth, hexagonal lattices behave differently. Luo and Zhang found that for spin-twisted hexagonal lattices "All twist angles $\theta \lesssim 6^\circ$ can become magic that support gapped flat bands", at a coupling that grows almost linearly with θ [23], and Sui *et al.* report for honeycomb lattices an isolated topological flat band at a critical coupling $\Omega_R \approx 0.11\,E_\mathrm{R}$ at $\theta \approx 4.4^\circ$, with trivial flat bands at strong coupling [24]. Fifth, for incommensurate twists the interlayer coupling acts as effective disorder, and above a critical value part of the spectrum becomes multifractal [22]. The original proposal of twisted state-dependent potentials [18] and a fermionic "twistronics without a twist" with moiré-modulated Raman coupling [35] complete the single-particle picture; a continuously rotated lattice acts as a position-dependent drive with ring-shaped localised states [29].

5.3 Lattice models with interactions

Paul, Recher and Santos wrote the two-layer Bose–Hubbard model with Gaussian interlayer hopping and density–density terms, $t_\perp(j,j') = t_\perp e^{-|\mathbf{R}_{1j}-\mathbf{R}_{2j'}|^2/4l_0^2}$, and treated it with cluster Gutzwiller after "site clusterization": pairs with coupling above $t_{\perp,\mathrm{cr}} = 0.03\,t_\perp$ (commensurate) or $0.06\,t_\perp$ (incommensurate) form clusters [21]. They find "Mott-like phases … resulting typically in a density wave modulation" and the "formation of mobility islands", where Bose-glass phases may occur. As the project documents summarise it, the runs use $t_\perp = U$, $V_\perp = 4U$ and $T = 0$; dynamics is left open. Zeng, Zhu and He removed the interlayer tunnelling altogether and let the layers couple only through contact interactions, at angles 36.87°, 22.62° and 7.63°; Gutzwiller calculations give induced Mott insulators, an "interlocked" Mott insulator "with spontaneously formed moiré lattice" and self-localised Bose-glass-like phases [25]. The arXiv versions carry two titles, "Interaction induced moiré systems" and "Dynamical Moiré Systems". A follow-up finds that interactions can delocalise a disconnected Bose glass into a connected superfluid at low filling, with reentrance at higher filling, and studies quenches [26]. For spin-1 condensates in a twisted trilayer, depth quenches create vortex pairs that inherit the moiré periodicity [30].

5.4 Continuous space: quantum Monte Carlo and GPE

Johnstone *et al.* ran continuous-space worm QMC for single-component bosons in the summed twisted potential (depths 3–6 E_R, $\theta \approx 28.07°$ and an incommensurate 27.13°, $k_BT = 0.01$–$0.04\,E_\mathrm{R}$). Density-wave and Mott plateaus are separated by thin superfluid domains with J_max between 10^{-11} and $10^{-3}\,E_\mathrm{R}$, and the incommensurate case leaves "a prominent Bose glass phase" [27]; a companion paper fits four forms of tight-binding model [28]. Real-time GPE work is scarce. Fan *et al.* solve the Meng-type coupled GPE (hexagonal lattices, $\Omega = 1\,E_\mathrm{R}$, $V_0 = 4\,E_\mathrm{R}$, $\theta = 9.43°$, $N = 2\times10^4$, trap $2\pi\times(20,20,1000)$ Hz) with a Floquet-modulated scattering length and obtain quasicrystal density patterns [31]; a preprint shows that the nonlinearity alone can self-organise a twisted-bilayer structure in a two-component BEC [79]. Gap solitons and vortices in moiré potentials fill a separate strand, with stability studied by linear or nonlinear methods on localised states, according to the abstracts [80–84].

5.5 One dimension and fermions

In 1-D, repulsion broadens the moiré flat band of a condensate in a $\mathcal{PT}$-symmetric bichromatic lattice [37], and a density-matrix renormalisation group (DMRG) preprint finds a "local supersolid" in a moiré-modulated Bose–Hubbard chain, with "A vanishing global structure factor in the thermodynamic limit, while the local structure factor remains finite" [38]. For fermions, Yang *et al.* (preprint, September 2026) take honeycomb lattices with uniform Ω and, as reported, a cross-lattice admixture $w \approx 0.05$; the single-particle spectrum "hosts a pair of quasi-degenerate nearly flat moiré bands", Hartree–Fock interactions "spontaneously break time-reversal symmetry" into a Chern band, and exact diagonalisation finds a fractional Chern insulator (FCI), reported at hole filling $\nu_h = 1/3$, with a many-body gap "of order $10^{-3}\,E_R$" [34].

5.6 What the coverage shows

Figure 5.1 makes the pattern plain. Most TBOL studies treat single-particle bands or ground states at $T = 0$; the finite-temperature work (Johnstone) is for a single-component summed potential; the dynamics (Fan, Ding, Li) concerns quenches of the scattering length, the interaction or the depth, not of Ω. The graphene literature and the superfluid-fraction experiments, placed in time in Figure 5.2, ran in parallel with the TBOL work [2, 3, 7, 10–12, 14, 49, 52, 85].

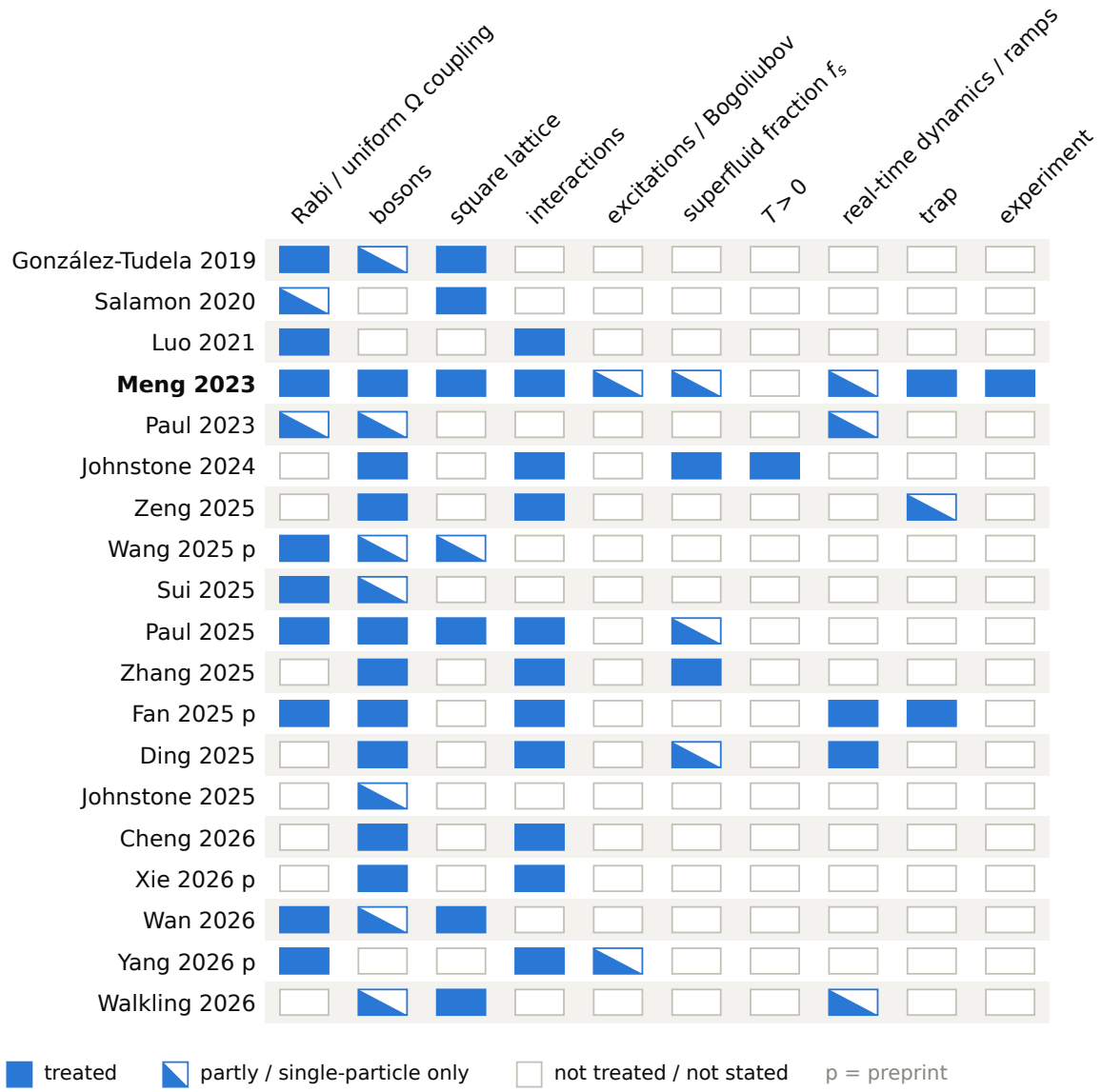


Figure 5.1: Coverage of twisted-bilayer optical-lattice theory and the experiment. Rows are studies (p: preprint), columns are ingredients from the coupling scheme to the trap. Filled: treated; half: partly treated or single-particle only; empty: not treated, or not stated in the abstracts and pages read for this map.

5.7 What is openly available

The worm QMC code for lattice bosons [65] is public under GPL-3.0, with 26 commits and no releases [86]. No TBOL data set has been released. The Nature paper states that "All data generated or analysed during this study are included in this published article", with further data "available from the corresponding authors upon reasonable request"; it has no code-availability statement [1]. The neural-quantum-state code for the 2-D Bose–Hubbard model of [68] is archived on Zenodo as v1.0.0 under the Apache-2.0 licence [87].

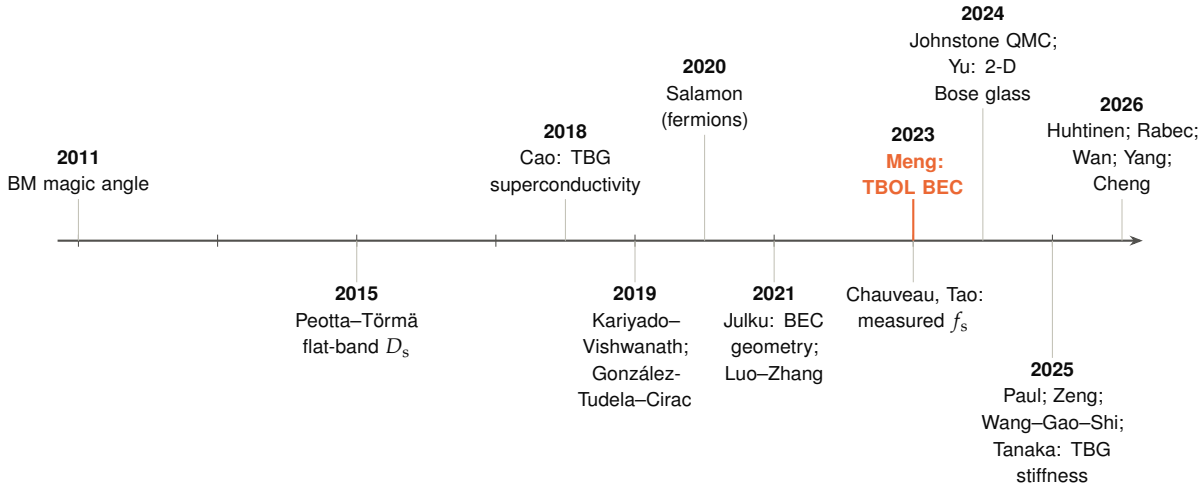


Figure 5.2: Timeline 2011–2026 of the work this map draws on: graphene and quantum geometry (2011–2021), the experiment (orange), superfluid-fraction measurements and TBOL theory (2023–2026). Salamon *et al.* and Yang *et al.* treat fermions.

Not found for this map

No calculation of the following was found in the literature checked for this map:

- Bogoliubov spectra, sound velocities or the superfluid-fraction tensor of the Rabi-coupled condensate;
- the quantum geometry of its bosonic bands (only fermionic honeycomb bands [34]);
- finite-temperature or truncated-Wigner behaviour of the Rabi-coupled model;
- ramps or quenches of Ω;
- an independent trapped simulation compared with the Nature data.

Key points

- One experiment: ^{87}Rb, 5.21°, $\Omega \leq 1\, E_R$, two superfluid regimes, no follow-up found [1].
- Uniform Ω acts as a scalar moiré potential; magic angles appear only in hexagonal lattices [5, 6, 23, 24].
- Interacting theory is Gutzwiller at $T = 0$, or QMC for single-component relatives with thin, fragile superfluid domains [21, 25, 27].
- No released data; the worm code is GPL-3.0 [86].

6 Position: geometric superfluidity is excluded; the pocket crossover is the open problem

Why a uniformly coupled square bilayer cannot hold a geometric flat-band superfluid, and why the experiment's second superfluid is best read as a Josephson array of AA pockets.

The claim of this chapter is limited to square twisted-bilayer optical lattices (TBOL) with a spatially uniform Rabi coupling at zero detuning, and it rests on single-particle band structures plus the published theory of condensates in flat bands; honeycomb lattices, modulated couplings

and detuned drives are outside it. We argue one thesis and hold a second as a working hypothesis with stated falsifiers. Figure 6.1 separates what is fixed from what is open, and Table 6.1 records how the framing of the project changed.

6.1 Thesis 1: the flat limit is the atomic limit

Symmetry. Kariyado and Vishwanath showed that flat bands in twisted Bravais bilayers, the square lattice included, arise from an interlayer coupling that varies in space, where symmetry forces the coupling matrix elements to cancel for specific momenta and displacements, "an interference effect where symmetry forces exact cancellation" [5]. A microwave Rabi coupling has no spatial structure, so this route is closed.

Effective theory. For a 3-D quasi-moiré crystal, Wang, Gao and Shi project the two-component Hamiltonian onto the lowest band and obtain a scalar moiré potential whose amplitude is proportional to Ω times a Gaussian form factor, valid for $\theta \cdot l_{\mathrm{ho}}^{-1} \ll 1$ [6] (preprint). Applied to the 2-D square bilayer, which is our step and not theirs, a uniform coupling becomes an ordinary potential landscape with minima in the AA registry regions.

Computation. The single-particle bands confirm this picture [computed for this map]. At $m = 8$ and $V_0 = 8\,E_{\mathrm{R}}$, the width of the lowest band relative to its uncoupled value is $W_1/W_1(0) = 0.85$, 0.66, 0.25, 0.043 and 0.004 at $\Omega/J = 6.5$, 9.7, 16, 23 and 32. The indirect gap to band 2 opens between $\Omega/J \approx 10$ and 16, and the mass ratio m/m^* at Γ falls from 0.28 to 0.003 over $\Omega/J = 3.2$–32. The band flattens by localisation: the coarse-grained AA/AB density ratio is about 110 at $\Omega/J \approx 16$, and at $\Omega/J \approx 32$ the Wannier spread is $0.022\,L_{\mathrm{M}}^2$ and the participation ratio 0.008. These numbers contain no interactions, trap or temperature, and the experiment's angle ($m = 22$) was not computed.

Theory. For condensates of uniform density, the preprint of Huhtinen *et al.* gives $D_1 + D_2^{\mathrm{ndeg}} = (2Un_0^2/M)\,\mathcal{M}^D(\mathbf{k}_c)$ with the condensate quantum metric at the condensation momentum, and states that "There can therefore be no superfluidity in a single flat band at a TRIM where the Bloch state has uniform densities."; stable two-dimensional flat-band superfluidity "requires at least three bands within Bogoliubov theory" [7]. The pocket band does not have uniform density, so this theorem does not apply literally. What excludes the geometric mechanism is the atomic limit. The review of Törmä, Peotta and Bernevig states that "a trivial flat band is equivalent to an insulator in the atomic limit of large atom-atom separation" and that there "no current is possible." [4]. The relevant metric in isolated flat bands is the minimal one, and a non-isolated band can do better [15]; for fermions, the stiffness of a trivial flat band is bounded by the Wannier spread [51], which here is about 2 % of L_{M}^2. Two further results point the same way: a two-band condensate has zero superfluid density "in the case of U(2) invariant interactions" [59] (preprint), and the experiment is modelled with SU(2)-symmetric $g_{11} = g_{22} = g_{12}$ [1]. We conclude that the geometric mechanism cannot carry D_{s} in the collapsed regime. No calculation of the condensate metric for the Rabi-coupled bands was found for this map, so the conclusion rests on the atomic-limit argument, not on a computed $\mathcal{M}^D$.

The window matters because the experiment sits in it. At $V_0 = 4\,E_{\mathrm{R}}$, where the moiré and superfluid data were taken, $\Omega_{\mathrm{R}} \leq 1\,E_{\mathrm{R}}$ corresponds to $\Omega/J \leq 12$ and $W_1 \geq 0.93\,W_1(0)$ at $m = 8$ [computed for this map]; the band is still dispersive. The threshold also moves with depth: at $V_0 = 12\,E_{\mathrm{R}}$ the band has already lost 96 % of its width at $\Omega/J \approx 16$, while at $4\,E_{\mathrm{R}}$ it halves only near $\Omega/J \approx 35$ [computed for this map].

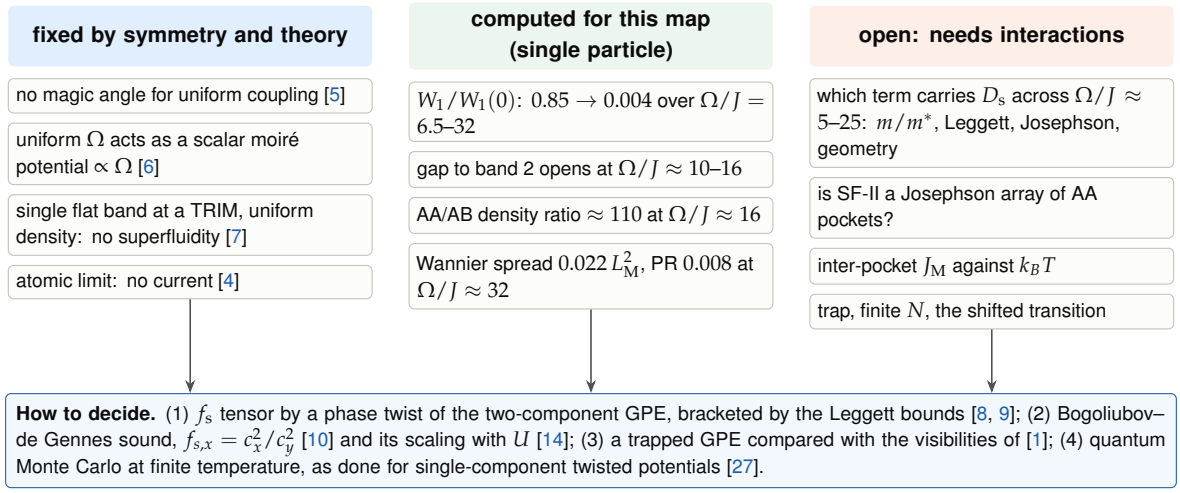


Figure 6.1: Position diagram. Left: results that hold independently of this map; middle: single-particle bands at $m = 8$, $V_0 = 8\,E_R$ (no interactions, trap or temperature; the experiment's $m = 22$ was not computed); right: what requires interacting calculations; bottom: the calculations and measurements that would decide.

Table 6.1: How the framing changed between the project's first proposal (v1) and this map (v2).

Question	v1 framing	v2 framing (this map)
how the band flattens	interference, as in TBG; a magic angle	localisation in AA pockets; no magic angle
control parameter	twist angle θ	Ω/J, with a V_0-dependent prefactor
what carries D_s	flat band plus quantum metric	competition of m/m^*, Leggett and Josephson terms across $\Omega/J \approx$ 5–25
role of geometry	the mechanism, centrepiece of the project	a diagnostic of the crossover
reading of SF-II	geometric flat-band superfluid	Josephson array of AA pockets (hypothesis)
central calculation	quantum metric of the flat band	f_s tensor by phase twist, Leggett brackets, sound

6.2 Thesis 2 (working hypothesis): SF-II is a Josephson array of AA pockets

The experiment describes its second superfluid regime as "superfluid domains embedded in a gapped insulator" [1]. We read these domains as the AA pockets of Thesis 1, with the AB regions insulating and phase coherence carried by tunnelling J_M between pockets. Three lattice and Monte Carlo studies found related textures. Paul, Recher and Santos find Bose-glass-like "mobility islands" in a two-layer Bose–Hubbard model with Gaussian-overlap interlayer hopping, even at commensurate angles [21]. For dipolar bosons in a twisted bilayer at 5.21°, "a periodic structure of insulating islands is formed. These insulating islands are surrounded by a superfluid." [36]: the same geometry with the roles exchanged. Continuous-space Monte Carlo of a single-component twisted potential finds density-wave plateaus separated by thin superfluid domains, with maximal tunnelling J_{max} between 10^{-11} and $10^{-3}\,E_R$ and thermal broadening that "strongly depletes the SF domains" [27].

The hypothesis has one quantitative anchor. The experiment writes E_r for the recoil energy; its moiré visibility at $V_0 = 8\,E_R$ "emerges at $\Omega_R > 0.5E_r$" [1]. At $m = 8$ this is $\Omega/J \approx 16$, where the lowest band detaches from band 2 and has fallen to $W_1 = 0.25\,W_1(0)$ [computed for this map]. The coincidence is suggestive, not a derivation: it compares an interacting, trapped measurement at $m = 22$ with non-interacting bands at $m = 8$.

Falsifiers. The hypothesis fails if any of the following is observed or computed.

F1 The superfluid fraction exceeds the Leggett upper bound evaluated on the pocket density: an array of weak links cannot beat it [8, 9].

F2 The sound velocity scales linearly with U, the signature of a geometric flat-band condensate [14], rather than as $\sqrt{U}$ with a Josephson-renormalised band mass [43].

F3 The visibility threshold does not track Ω/J as V_0 changes. The experiment reports its 0.5 E_R threshold for path III, run at 8 or 14 E_R [1]; extrapolating the grid, the single-particle collapse at 14 E_R would already sit near 0.15–0.2 E_R [computed for this map] (no grid point at 14 E_R), so either interactions or temperature shift it, or the pocket reading is wrong.

F4 The inter-pocket J_M at the relevant Ω lies far below any plausible k_BT, on the scale of the Monte Carlo J_{max} [27]; SF-II would then be a finite-size or interaction-restored effect, not a thermodynamic Josephson array. The experiment does not state its temperature.

Where quantum geometry still enters

Geometry stays in the problem as a diagnostic. The bare metric at Γ peaks at about 0.027 L_M^2 as the band collapses and decays beyond it [computed for this map], and the condensate metric at $\mathbf{k}_c$ sets sound and stability in the bosonic theory [7, 14]. A genuinely geometric window could open at band touchings, where "a non-isolated flat band can actually be beneficial" [15], and under detuning, which maps onto "Hofstadter's original model plus a long-range hopping term" [16]. Honeycomb bilayers are different: uniform coupling there gives "magic" isolated flat bands for all $\theta \lesssim 6°$ [23] and an isolated topological flat band at a weak critical coupling [24].

Key points

- Uniform Ω on a square bilayer gives no magic angle; the band flattens by AA localisation [5, 6].
- In the collapsed regime the band is in the atomic limit, where no geometric superfluid weight survives [4, 7].
- The open physics is the crossover $\Omega/J \approx 5$–25, where the experiment operates.
- Working hypothesis: SF-II is a Josephson array of AA pockets; four falsifiers are stated.

7 Around the topic

Neighbouring systems that share one ingredient with the twisted-bilayer condensate: moiré patterns for light, quasiperiodic lattices, dipolar bilayers, coupled 1-D condensates and moiré magnets.

None of these systems is a Rabi-coupled moiré Bose gas; each offers a tool or a measurement to borrow.

7.1 Photonic moiré lattices

Light in a photonic moiré lattice obeys, in the paraxial approximation, an equation of the same form as the single-particle Schrödinger equation, and with a Kerr nonlinearity the form of the GPE. In such lattices Wang *et al.* observed "the two-dimensional localization–delocalization transition of light" [32]. The same group showed that soliton formation is controlled by the twist angle, with "threshold properties that are a pristine direct manifestation of flat-band

physics" [33]. In nanostructured moiré superlattices, "the confinement mechanism of magic-angle lasers does not rely on a full bandgap but on the mode coupling between two twisted layers" [88]. Reviews cover moiré photonics and optoelectronics [89] and dielectric photonic moiré metamaterials [90]; a 2024 editorial surveys moiré physics beyond van der Waals materials [91], and the project documents cite a 2024 review of photonic and polaritonic moiré whose identity was not confirmed [92]. The closest analogue to our problem is condensation itself: exciton polaritons condense "in a perovskite moiré flat band at room temperature" [93] (title; authors not obtained for this map). Being driven and dissipative, they do not test the equilibrium superfluid fraction.

7.2 Optical quasicrystals and the 2-D Bose glass

The incommensurate limit of a twisted lattice is a quasiperiodic potential. Yu *et al.* reported "the experimental realization of the two-dimensional Bose glass using ultracold atoms in an eight-fold symmetric quasicrystalline optical lattice" [85]. A 2026 preprint from the same group quenches across the Bose-glass transition and finds that the characteristic timescales "are governed primarily by quasi-disorder strength in the Bose glass phase, and by tunneling strength in the superfluid phase" [94]. Both are single-component and without a Rabi coupling, but they show that a Bose glass in 2-D is measurable, which matters if SF-II is to be told apart from one.

7.3 Dipolar twisted bilayers

Zhang, Fan, Capogrosso-Sansone and Deng studied hard-core dipolar bosons in two twisted layers where "particles cannot hop between layers", with path-integral QMC using the two-worm algorithm and the superfluid stiffness as observable [36]. At the experiment's angle of 5.21° "a periodic structure of insulating islands is formed. These insulating islands are surrounded by a superfluid." The layers couple only through the long-range interaction, as in the contact-coupled model of [25], and the resulting picture of insulating pockets in a superfluid network is the inverse of the superfluid pockets in an insulator that SF-II suggests.

7.4 Coupled one-dimensional condensates in Vienna

The relative phase of two coupled condensates is the one-dimensional cousin of the interlayer phase in a TBOL. On an atom chip, "a pair of tunnel-coupled one-dimensional atomic superfluids" in thermal equilibrium "can be seen as a quantum simulator of the sine-Gordon model" [95]. With Raman tunnel coupling, the commensurate–incommensurate transition "can be … described by the Pokrovsky-Talapov model for the relative phase", with "a soliton staircase in the phase difference" [96]. Mennemann *et al.* studied relaxation in this extended bosonic Josephson junction: "stochastic classical field simulations using Gross-Pitaevskii equations in three spatial dimensions reproduce the main experimental findings", the relaxation persists "even at zero temperature", and it could not be explained by the sine-Gordon model [97]. For coherently (Rabi) coupled condensates the spin mode is gapped by the coupling, and "finite spin-flipping coupling changes the first order demixing phase transition … to a second order" [39]; experiments studied ferromagnetism in an extended coherently coupled superfluid [98] and observed "bubble nucleation in isolated and coherently coupled atomic superfluids", a false-vacuum decay [99]. These are the same two ingredients as in a TBOL, a Rabi coupling and a spatially varying relative phase, without the lattices.

7.5 Magnetic moiré

Twisted magnets form moiré patterns in the spin texture. A 2025 review covers twisted CrI_3 with "moiré magnetism in nanoscale domains and periodic magnetization patterns", "skyrmion lattices and helical or spin spirals", and "moiré magnon excitations" [100]. Moiré magnons are the closest analogue of the Bogoliubov modes of a moiré condensate, for which no spectrum was found for this map.

Key points

- Photonic moiré lattices share the GPE mathematics and show a localisation–delocalisation transition and flat-band soliton thresholds [32, 33].
- The 2-D Bose glass has been observed in an optical quasicrystal, the incommensurate relative of a TBOL [85, 94].
- Dipolar twisted bilayers at 5.21° give insulating islands in a superfluid [36].
- Coupled 1-D condensates show that relative-phase dynamics needs 3-D classical-field simulation, not sine-Gordon alone [97].

8 Research and industry perspective

Who works on twisted optical lattices and flat-band superfluids, which codes and budgets exist, and where the neighbouring superconductivity research stands in 2026.

This chapter reports sourced facts only: group pages, code repositories, programme and company statements, and the published record. Versions and licences are as stated on the pages, accessed on 21 September 2026. Statements by companies are marked "according to the company".

8.1 Platforms and groups

One laboratory has realised a twisted-bilayer optical lattice (TBOL) with a Bose gas, the Shanxi–Chicago collaboration of Jing Zhang and Cheng Chin [1]; its later work is on untwisted and twisted honeycomb lattices [20, 24]. The superfluid-fraction measurements a TBOL study would copy come from three groups that have not worked on twisted lattices (Table 8.1). The Santos group page lists one twisted-bilayer paper [101]; the affiliation of the fermionic proposal [34] was not checked.

8.2 Open codes and data

No TBOL data set has been released. The experiment's data statement reads "All data generated or analysed during this study are included in this published article." [1]. The codes a TBOL study would draw on are listed in Table 8.2. The worm quantum Monte Carlo (QMC) code builds on ALPSCore and shows "near-linear scaling" in volume and inverse temperature [65]. None of the Gross–Pitaevskii equation (GPE) codes offers the Rabi-coupled twisted lattice as a ready model, as far as the pages state.

8.3 Compute and cost, as quoted

For the Kibble–Zurek problem in the 2-D Bose–Hubbard model, infinite projected entangled-pair states need bond dimension $D = 11$–14, and "The dominant numerical cost of the NTU

Table 8.1: Groups active on the topic or on the methods it needs, with the papers cited in this map. Affiliations of the Palaiseau, Paris, Cambridge and Vienna groups follow the project documents and were not re-checked.

Group	Work relevant here	Papers
Shanxi / Chicago: J. Zhang, C. Chin, Z. Meng	the TBOL experiment; honeycomb follow-ups	[1, 19, 20, 24]
Palaiseau (CPHT): L. Sanchez-Palencia	continuous-space QMC of twisted potentials	[27, 28, 45]
Hannover: L. Santos	TBOL lattice models, cluster Gutzwiller	[21, 22]
Aalto: P. Törmä	flat-band superfluid weight, quantum geometry	[4, 7, 14, 49]
ETH Zürich: S. D. Huber	Bose condensation in flat bands	[102]
Paris (Collège de France): J. Dalibard, J. Beugnon	f_s from Leggett bounds and sound	[10, 12]
NIST / JQI: I. B. Spielman	f_s from Bragg phonons in a 1-D lattice	[11]
Florence (CNR-INO): G. Modugno, with M. Fattori	f_s from Josephson oscillations of a supersolid	[13]
Cambridge: U. Schneider	2-D Bose glass in an optical quasicrystal; quenches	[85, 94]
Vienna (TU Wien, Atominstitut): J. Schmiedmayer	tunnel-coupled 1-D condensates	[95–97]
Y. Xu and co-workers	fermionic TBOL, fractional Chern insulators	[34]

Table 8.2: Open codes, with licence and version as stated on the pages (accessed 21 September 2026).

Code	Use here	Licence	Version, status
worm QMC [65, 86]	lattice bosons, ρ_s from windings	GPL-3.0	no releases
ALPSCore [103]	QMC libraries	MIT	v2.3.3, May 2026
NetKet [104]	neural quantum states	Apache-2.0	3.22.4, Aug 2026
TeNPy [105]	tensor networks	Apache-2.0	v1.1.0, Nov 2025
ITensors.jl [106]	tensor networks	Apache-2.0	v0.9.30
lattice-bosons [68, 87]	neural states, Bose–Hubbard	not checked	v1.0.0, June 2025 (Zenodo)
XMDS2 [107]	stochastic PDEs, TWA	GPL-2.0	docs 2.2.2; news ends 2014
GPELab [108]	MATLAB GPE, stochastic GPE	none stated	none stated
GPUE [109]	CUDA GPE	not checked	JOSS 2018
GPU Python GPE [110]	PyTorch GPE, cavity QED	not checked	preprint 2024
quTARANG [111]	HPC GPE turbulence	MIT	CPC 2025
FourierGPE.jl [112]	FFT GPE	MIT	"no longer actively maintained"
GeneralizedGross-Pitaevskii.jl [113]	stochastic, GPU GPE	MIT	not checked
Wannier90 [114]	maximally localised Wannier functions	LGPL-2.1	v3.1.0, Feb 2024

procedure scales as D^8" [78]. A spin-1 stochastic projected GPE study uses, as reported, 2.5×10^5 steady-state samples on a 512×512 grid [115]. The cluster truncated Wigner method runs 2×2 clusters on 20×20 sites [75], and a neural backflow ansatz reaches lattices up to 20×20 [68]. The continuous-space QMC of twisted potentials uses boxes of 5 and 15 moiré cells at $k_B T = 0.01$–$0.04\, E_R$ and gives no cost statement [27]. The project's band code took 7–35 s per (V_0, Ω) point at $m = 8$ and 15–108 s at $m = 12$ on 2 CPU cores [computed for this map]. No timing figure for a GPU GPE solver was found for this map.

8.4 Programmes and prizes

The EU Quantum Flagship project PASQuanS2 was launched in April 2023 for seven years, with a first phase of 3.5 years, coordinated by Immanuel Bloch; it aims at "quantum simulators with at least 2,000 atoms and a path towards 10,000"; the EU amount is not on the page [116]. The Austrian cluster of excellence quantA started on 1 December 2023 with "up to 70 million euros" over ten years for six partners, including TU Wien and the University of Vienna [117]. In the US, the NSF Quantum Leap Challenge Institutes round of 25 August 2026 gives more than $290 M to eight institutes over five years, "between about $28 and $37 million" each [118]; the first award to the Robust Quantum Simulation institute was $25 M in 2021 [119]. The ERC Consolidator grant QTWIST (Manchester, 1 April 2021 to 31 March 2027) receives €2,880,313 for twistronics; the PI is not on the page [120]. No ERC grant on superfluidity of quantum gases in moiré lattices was found for this map. The field was recognised in 2026: the Kavli Prize in Nanoscience went to Eva Andrei, Pablo Jarillo-Herrero and Allan MacDonald "for foundational work that established the field of Twistronics", worth USD 1,000,000 [121].

8.5 Companies

According to the company, planqc builds the MAQCS system for the Leibniz Supercomputing Centre in Garching, a €20 M project funded by the German research ministry (BMBF), with "two independent cores of 1,000 qubits in total" that "will employ neutral atoms in optical lattices as qubits" [122]. QuEra's Aquila, a "field-programmable qubit array" in the authors' words, offers analog simulation on up to 256 neutral-atom qubits through Amazon Braket [123] (preprint). Lattice quantum simulation and atomtronics are reviewed in [124, 125]; no company offering moiré simulation, and no substantive commercialisation of twistronics, was found for this map.

8.6 Flat-band and quantum-geometric superconductivity

The electronic side of the question is active. Magic-angle graphene superconducts at twist angles of about 1.1° with "a critical temperature of up to 1.7 kelvin" [3]. Its superfluid stiffness, measured through the kinetic inductance, is "much larger than expected from conventional Fermi liquid theory" [52], with a companion measurement in twisted trilayer graphene [53]; an earlier circuit measurement [126] is described in [55] as "approximately ten times larger than a conventional BCS estimate". The single-author preprint [55] attributes 22–26 % of D_s at charge neutrality to quantum geometry within the flat bands and about 55–58 % once remote bands are included (model calculation). Twisted WSe_2 superconducts at 3.5° and 3.65° with an "optimal superconducting transition temperature [that] is about 200 mK in both cases" [127], at 5.0° with a "maximum critical temperature of 426 mK" [128], and "evolves smoothly with twist angle" in between [129]. Rhombohedral tetralayer and pentalayer graphene, "without moiré superlattice effects", shows chiral superconductivity with "Tc up to 300 mK" at densities down to 2.4×10^{11} cm^{-2} [130]; aligned to hBN, it hosts a quantised anomalous Hall state, a superconductor and a fractional Chern insulator at zero field [131]. Fractional quantum anomalous Hall states appear in twisted $MoTe_2$ without a magnetic field [132], with "nearly 20 hidden states at fractional fillings" at 3.1°, 3.7° and 5.5° [133]; a Perspective compares the two platforms [134], and theory predicts a non-Abelian fractional Chern insulator in the half-filled second miniband [135]. A kagome metal under pressure adds a non-moiré flat-band superconductor [136]. The quantum-geometric view is collected in a review of superfluidity [137], a Perspective on geometry in materials [138] and a review in National Science Review whose authors were not read for this map [139].

What the cold-atom platform offers this research

No twist-angle disorder, strain or Coulomb interaction; Ω and V_0 tuned in one run; and a superfluid fraction measured directly, from sound and Leggett bounds [10, 12], Bragg phonons [11] or Josephson oscillations [13]. It tests the bosonic side, where geometric superfluidity is predicted to be harder to reach [7].

Key points

- One experimental group, several theory groups; the f_s measurements come from three groups outside TBOL.
- Open QMC, tensor-network and GPE codes exist; no TBOL data set or TBOL-ready GPE code was found.
- Funding reaches quantum simulation broadly (PASQuanS2, quantA, NSF QLCI); twistronics has an ERC grant and the 2026 Kavli Prize.
- Electronic flat-band superconductivity supplies the questions: stiffness, geometry share, angle dependence.

9 Open questions

Twenty-two questions in five themes, from effective models of the pockets to finite temperature and geometry, followed by directions that no study checked for this map has taken up.

The questions follow from Chapter 6; Figure 9.1 shows them at a glance. "Not found" refers to the literature checked for this map (cutoff 21 September 2026).

models	**Q1** pocket Wannier functions, J_M, U_M	**Q2** scalar potential at finite θ	**Q3** detuning with interactions	**Q4** incommensurate angles	**Q5** the experiment's $m = 22$
f_s and sound	**Q6** f_s tensor by phase twist	**Q7** Leggett brackets vs metric	**Q8** sound anisotropy, U scaling	**Q9** depletion in pockets	**Q10** relative-phase branch
experiment	**Q11** SF-II: Josephson array or Bose glass	**Q12** the Ω-shifted transition	**Q13** visibility threshold vs V_0	**Q14** trap, N, scattering lengths	
finite T, dynamics	**Q15** J_M vs k_BT; BKT of pockets	**Q16** Ω ramps, defect counts	**Q17** TWA at low filling	**Q18** extended-junction relaxation	
geometry	**Q19** condensate metric of Rabi bands	**Q20** a genuine geometric window	**Q21** bosonic fractional Chern states	**Q22** fermions, dipoles, photonics	

Figure 9.1: The 22 open questions, one row per theme: single-particle theory and effective models; superfluid fraction and excitations; the experiment's phases; finite temperature and dynamics; geometry and beyond.

9.1 Single-particle theory and effective models

Q1 **Pocket Wannier functions.** What are the Wannier functions of the AA pockets, and how do the inter-pocket tunnelling $J_M(\Omega)$ and on-pocket interaction $U_M(\Omega)$ depend on the coupling? Fitted tight-binding models exist for single-component twisted potentials [28], and Wannier tooling exists [114], but not for the Rabi-coupled bands.

Q2 **The scalar potential at finite angle.** How accurate is the scalar moiré potential, derived for $\theta \cdot l_{\mathrm{ho}}^{-1} \ll 1$ in three dimensions [6], for the 2-D square bilayer? The Gaussian-overlap interlayer coupling [18, 21] underestimates the bandwidth by 30 % at $\Omega/J \approx 10$ and tenfold at $\Omega/J \approx 32$ [computed for this map].

Q3 **Detuning with interactions.** The detuned model maps onto "Hofstadter's original model plus a long-range hopping term" [16] for single particles; how do interactions act there, where Gutzwiller studies of quasicrystal limits find interaction-induced reentrance of the Bose glass [26]?

Q4 **Incommensurate angles.** Incommensurate twists act as effective disorder with multifractal states [22], and continuous rotation gives ring-shaped localised states [29]; the Rabi-coupled continuum at incommensurate angles with interactions was not found beyond Gutzwiller.

Q5 **The experiment's angle.** The bands were computed for $m \leq 12$, and at $V_0 = 4\,E_{\mathrm{R}}$, $\Omega = 1\,E_{\mathrm{R}}$ the ratio $W_1/W_1(0)$ drops from 0.93 at $m = 8$ to 0.59 at $m = 12$ [computed for this map]; the experiment's $m = 22$ was not computed.

9.2 Superfluid fraction and excitations

Q6 **The f_{s} tensor.** What is the superfluid-fraction tensor [40] of the two-component condensate under a phase twist [41] across $\Omega/J \approx 5$–25? It has been measured in full for a triangular lattice [12], not for a moiré lattice.

Q7 **Leggett brackets and the metric.** No paper linking the Leggett bounds [8, 9] to the band quantum metric for bosons was found, as the project documents state and the checks did not contradict; angular dependence in composite potentials is a first step [42].

Q8 **Sound.** Does the Bogoliubov–de Gennes sound anisotropy $f_{s,x} = c_x^2/c_y^2$ [10] follow the Bloch-band mass [43], or does it turn linear in U as in flat bands [14, 60]?

Q9 **Depletion.** In a geometric flat band the depletion stays finite as $U \to 0$ [14]; how large is it in the pockets, and do the Bogoliubov modes carry topology, as in some flat-band condensates [140]?

Q10 **The relative-phase branch.** Rabi coupling turns the demixing transition from first to second order [39]; with a moiré-modulated density the balance of coupling and interaction varies in space. Its effect on ferromagnetic order [98] and bubble nucleation [99] in a twisted lattice is open.

9.3 The experiment's phases

Q11 **The nature of SF-II.** Is SF-II [1] a Josephson array of AA pockets or a Bose glass? Lattice models find "mobility islands" [21], induced Mott phases [25] and, for dipoles, insulating islands in a superfluid [36].

Q12 **The shifted transition.** Can a continuum calculation reproduce the Ω-dependent superfluid–insulator line? Single-band Bose–Hubbard "fails to describe but the first Mott lobe" at 15 E_{R} in hexagonal potentials [45]; cluster Gutzwiller reaches QMC accuracy for square lattices [66, 67].

Q13 **Threshold and depth.** Does the 0.5 E_{R} visibility threshold [1] track Ω/J between 8 and 14 E_{R}, as the pocket picture predicts (Chapter 6)?

Q14 **Trap, atom number and scattering lengths.** As read for this map, the paper states neither the radial trap frequency nor the pancake atom number nor the temperature [1]; one GPE study assumes a $2\pi \times 20\,\mathrm{Hz}$ radial trap [31]. Scattering lengths are known for other hyperfine pairs [46, 47], but none for $|1,1\rangle/|2,0\rangle$ was found.

9.4 Finite temperature and dynamics

Q15 **Tunnelling against temperature.** Is J_{M} above $k_B T$? In single-component twisted potentials $J_{\max}$ spans 10^{-11}–$10^{-3}\,E_{\mathrm{R}}$ and temperature depletes the superfluid domains [27]. The Berezinskii–Kosterlitz–Thouless (BKT) transition of the pocket array would follow from Nelson–Kosterlitz [141] and the 2-D critical density [142], with stochastic GPE protocols such as the spin-1 study [115] as a template.

Q16 **Ramps of Ω.** Ramping Ω across a crossover, not a transition, will not give clean Kibble–Zurek exponents; lattice ramps already deviate [76, 77], and iPEPS needs large bond dimension [78]. Depth quenches produce moiré-periodic vortex pairs [30].

Q17 **Truncated Wigner at low filling.** Truncated Wigner needs fewer modes than particles [70, 72] and highly occupied modes [71]; SU(3) and cluster variants show "noticeable deviations from the experimental results even at relatively short times" in a strongly interacting 2-D quench [75]. Does a sparsely filled pocket array break it?

Q18 **The extended junction.** Two tunnel-coupled 1-D gases realise the sine-Gordon model [95], a Pokrovsky–Talapov transition [96], and relaxation to phase locking [97], with learned representations from snapshots [143]; the 2-D twisted analogue is open.

9.5 Geometry and beyond

Q19 **The condensate metric.** What is $\mathcal{M}^D(\mathbf{k}_c)$ [7] of the Rabi-coupled bands, with the orbital-independent metric [57], and does the U(2) result [59] transfer to non-uniform pockets? No calculation was found.

Q20 **A geometric window.** Band touchings [15], triangle frameworks [58], honeycomb magic couplings [23, 24] and Raman modulation (proposed for fermions) [35] are candidates for a regime where geometry carries D_{s}; which is reachable with bosons? Kagome and Lieb lattices for bosons have been realised [144–146].

Q21 **Bosonic fractional Chern states.** Interactions break time reversal and give a Chern band for fermions in a TBOL [34]; a bosonic analogue was not found.

Q22 **Transfer.** Which results carry over to fermions [23], dipolar bilayers [36] and photonic moiré lattices [32, 33], where polaritons condense in a moiré flat band [93]?

9.6 Directions not yet systematically addressed

As far as the checks for this map could establish, the following have not been taken up for the Rabi-coupled twisted Bose gas.

D1 Bogoliubov spectra of the extended condensate; the abstracts of the gap-soliton studies report none [80–83].

D2 Superfluid fraction with Leggett brackets, and finite temperature or truncated Wigner, for the two-component Ω-coupled model.

D3 Ramps or quenches of Ω; quenches exist for U, depth and $g(t)$ [26, 30, 31].

D4 An independent trapped simulation compared with the published data [1].

D5 A single account joining the continuum potential [6], the Hofstadter limit [16] and the cluster picture [21].

D6 2-D extensions of 1-D moiré results: nonlinear flat bands [37], local supersolids [38] and self-organised twisted bilayers [79].

D7 New f_s probes on moiré lattices: Josephson oscillations [13], sound in supersolids [63] and vortex necklaces [62].

D8 Neural-network surrogates for the TBOL GPE: physics-informed networks [147], deep ground-state solvers [148] and GPE applications [149, 150], with constrained variants from micromagnetics [151, 152].

Key points

- The effective model of the pockets (J_M, U_M) is the first missing piece.
- The f_s tensor, sound and depletion of the Rabi-coupled condensate have not been computed.
- The experiment's missing parameters (trap, temperature, scattering lengths) limit any comparison.
- Geometry may still matter at band touchings, under detuning and in honeycomb bilayers.

10 Reading paths and outlook

Where to start reading, depending on the field one comes from, and what the next experiment or the first superfluid-fraction calculation would change.

10.1 Where to start

A reader from Bose-gas theory should begin with the experiment [1] and the reason its band has no magic angle [5, 6]. Then read the flat-band condensate of Julku, Bruun and Törmä [14], which shows what a geometric superfluid looks like (sound linear in U), and the preprint of Huhtinen *et al.* [7], which states when it cannot exist. The magic-angle paper of Bistritzer and MacDonald [2] supplies the vocabulary; chapter 3 is the short version.

For a condensed-matter moiré physicist the new ingredients are the superfluid fraction of a modulated condensate and how cold-atom experiments measure it. Start with Leggett's bound [8] and its two-dimensional brackets [9], then the sound-anisotropy measurement [10]. Read the experiment [1] next, and then the lattice model [21] and the continuous-space Monte Carlo study whose thin superfluid domains are depleted at accessible temperatures [27].

For numerical analysts and applied mathematicians the model is Eq. (1) on a commensurate supercell (chapter 2). For effective models with controlled errors, start with the derivation of the Bistritzer–MacDonald model as the effective dynamics of wave packets "with rigorously estimated error bounds" [153] and with the scalar moiré potential and its validity condition [6]. The detuned limit maps onto a Hofstadter model with a fractal spectrum [16]. Continuum and lattice descriptions are compared in [45], and an open worm-algorithm code for lattice quantum Monte Carlo is described in [65].

10.2 Outlook

The field rests on one experiment at one angle. A second experiment would matter most if it changed the lattice depth along the coupling path: if the moiré-visibility threshold then moves with Ω/J as the band-structure numbers predict, pocket formation is the mechanism; if it stays at a fixed coupling, interactions or temperature set it. A measured temperature and a sound-anisotropy or Josephson measurement of f_s, as already done in simpler modulated gases [10–13], would turn the phase labels SF and SF-II into numbers.

On the theory side, no superfluid-fraction calculation for the Rabi-coupled condensate was found in the literature checked for this map. A first one, a phase twist of Eq. (1) on the $m = 8$ supercell with Leggett brackets across $\Omega/J \approx 5$–25, would show which contribution carries f_s in the crossover: band mass, density inhomogeneity or tunnelling between pockets. It would also test the working hypothesis of chapter 6, that SF-II is a Josephson array of AA pockets; the experiment's $m = 22$, a trap and finite temperature would then meet the data.

Key points

- Start from the experiment, then flat-band condensates, Leggett's bound or the effective models, depending on background.
- Next experiment: vary the depth along the coupling path; measure f_s and the temperature.
- Next calculation: f_s of the Rabi-coupled condensate across the crossover, with Leggett brackets.

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A Glossary

Terms from cold-atom physics, electronic moiré materials and quantum geometry, as this map uses them.

Moiré pattern. The long-wavelength beat between two periodic structures that are rotated or stretched against each other; its period L_M is much larger than the lattice constant at small twist.

Commensurate supercell. The common superlattice of two lattices at a commensurate twist, for square lattices $\tan(\theta/2) = 1/m$ with integer m; it makes band calculations periodic.

Registry (AA, AB). How the sites of the two layers line up locally: on top of each other in AA regions, alternating in AB regions. AA regions become the pockets where atoms collect.

Rabi coupling. A coherent drive (microwave or two-photon Raman) that converts atoms between two internal states; in a twisted-bilayer optical lattice it plays the role of interlayer tunnelling. Written Ω.

Tune-out wavelength. A laser wavelength at which the light shift of one atomic state vanishes, so that a lattice at that wavelength traps only the other state; used to give each spin state its own lattice.

Recoil energy. $E_R = \hbar^2 k^2/2m$ for the lattice wavevector k; the natural energy unit of optical lattices. Lattice depths V_0 are quoted in E_R.

Bose–Hubbard model. Lattice model of bosons with nearest-neighbour hopping J and on-site repulsion U; valid when the lowest band is deep and well separated.

Gutzwiller approximation. Mean-field product of single-site states; captures the Mott transition but no inter-site correlations.

Cluster Gutzwiller. Gutzwiller theory with small clusters of sites treated exactly, which restores short-range correlations.

Worm quantum Monte Carlo. Path-integral Monte Carlo that samples open world-line ends; exact for bosons without a sign problem and gives the superfluid fraction from winding numbers.

TWA and SPGPE. Truncated Wigner approximation: classical fields with quantum noise in the initial state. Stochastic projected GPE: a classical-field theory coupled to a thermal reservoir.

GPE. Gross–Pitaevskii equation: the mean-field equation for the condensate wavefunction; here with two components and a Rabi term.

BdG (Bogoliubov–de Gennes). Linearisation around a mean-field state; for a condensate it gives the Bogoliubov excitation spectrum, sound velocity and depletion.

Superfluid fraction. f_s: the part of the fluid that does not follow a slowly moving boundary, or equivalently the response to a phase twist; a tensor in anisotropic potentials.

Superfluid weight. D_s: the coefficient of the energy cost of a phase twist (superfluid stiffness); f_s is D_s normalised by its free-particle value.

Leggett bounds. Upper and lower bounds on f_s computed from the density profile alone; the upper bound is exact for a one-dimensional modulation of a weakly interacting condensate.

Quantum metric. $g_{\mu\nu}(\mathbf{k})$: the real part of the quantum geometric tensor; it measures how fast the Bloch state changes with $\mathbf{k}$.

Condensate quantum metric. The metric evaluated at the condensation momentum $\mathbf{k}_c$ in the basis of the condensate; it sets sound and stability of a flat-band condensate.

Minimal quantum metric. The integrated metric in the orbital basis that makes its diagonal components smallest; the quantity tied to the superfluid weight of paired fermions in an isolated flat band.

Quantum distance. $1 - |\langle u_{\mathbf{k}} | u_{\mathbf{k}'} \rangle|^2$ between two Bloch states; its small-step limit is the quantum metric.

Wannier function and spread. A localised orbital built from the Bloch states of a band; the spread is its mean-square size. A small spread means a band of nearly isolated orbitals.

Participation ratio. A measure of how many sites or cells a density spreads over, normalised to one for a uniform density; it falls towards zero as atoms collect in pockets.

TRIM. Time-reversal-invariant momentum: a point of the Brillouin zone equal to its negative up to a reciprocal-lattice vector (Γ, X, M in a square lattice).

BM model. Bistritzer–MacDonald continuum model of twisted bilayer graphene: Dirac cones of the two layers coupled by moiré-periodic tunnelling.

Magic angle. A twist angle at which the Dirac velocity of the BM model vanishes and the lowest moiré band flattens; the first lies near 1.05° [2].

Flat band. A Bloch band whose width is small compared with the interaction energy, so that interactions decide the ground state.

FCI / FQAH. Fractional Chern insulator, fractional quantum anomalous Hall state: fractional quantum Hall physics in a flat Chern band without an external magnetic field.

Bose glass. An insulating but compressible phase of bosons in a disordered or quasiperiodic potential, with local superfluid puddles but no global phase coherence.

Josephson array. A lattice of superfluid islands coupled by tunnelling; its superfluid weight is set by the inter-island Josephson coupling.